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To cite this article: Manzhu Yu, Qunying Huang & Zhenlong Li (2023) Deep learning for spatiotemporal forecasting in Earth system science: a review, International Journal of Digital Earth, 17:1, 2391952, DOI: [10.1080/17538947.2024.2391952](https://doi.org/10.1080/17538947.2024.2391952)

To link to this article: <https://doi.org/10.1080/17538947.2024.2391952>



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Published online: 19 Aug 2024.



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Deep learning for spatiotemporal forecasting in Earth system science: a review

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ABSTRACT

Deep learning (DL) has demonstrated strong potential in addressing key challenges in spatiotemporal forecasting across various Earth system science (ESS) domains. This review examines 69 studies applying DL to forecasting tasks within climate modeling and weather prediction, disaster management, air quality modeling, hydrological modeling, renewable energy forecasting, oceanography, and environmental monitoring. We summarize commonly used DL architectures for spatiotemporal forecasting in ESS, key technical innovations, and the latest advancements in spatiotemporal predictive applications. While DL architectures have proven capable of handling spatiotemporal data, challenges remain in tackling the complexities specific to ESS, such as complex spatiotemporal data, scale dependencies, model interpretability, and integration of physical knowledge. Recent innovations demonstrate growing efforts to integrate physical knowledge, improve model explainability, adapt DL architectures for domain-specific needs, and quantify uncertainties. Finally, this review highlights key future directions, including (1) developing more interpretable hybrid models that synergize DL and traditional physical approaches, (2) extending model generalizability through techniques like domain adaptation and transfer learning, and (3) advancing methods for uncertainty quantification and missing data handling.

ARTICLE HISTORY

Received 1 May 2024

Accepted 8 August 2024

KEYWORDS

Deep learning;
spatiotemporal forecasting;
Earth system science; review

1. Introduction

Spatiotemporal forecasting is a critical task in Earth system science (ESS) involving the prediction of Earth phenomena, including weather patterns, climate change, geophysical processes, and ecosystem dynamics. Deep learning (DL) has become a transformative tool in ESS as it can learn from and make predictions based on complex and heterogeneous data. In particular, DL models are capable of effectively incorporating both spatial and temporal information, ensuring that data do not exist in isolation but are part of a broader, interconnected system. This advancement enables the prediction of ESS phenomena at any given point of a system and its propagation over space and time (Tascikaraoglu 2018).

Despite promising advancements and the availability of numerous strategies for enhancing spatiotemporal forecasting through DL, two primary challenges persist. One general challenge lies in the development of spatially explicit DL models that can handle the complex and implicit

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relationships of spatiotemporal objects, the modifiable areal unit problem (MAUP), the integration of heterogeneous datasets, and the issues related to spatiotemporal discretization (Hamdi et al. 2022; Lavallin and Downs 2021; Liu and Biljecki 2022). The second general challenge remains in physical explainability or consistency in DL models by integrating physical and DL models in both directions (Reichstein et al. 2019; Xu et al. 2021). Strategies include using DL to emulate physical models, calibrating physical model outputs, and designing DL architectures that respect physical laws (Yuan et al. 2020). Kashinath et al. (2021) highlighted the critical challenges in weather and climate modeling, including capturing complex multiscale physical processes and spatiotemporal dynamics. Overcoming these hurdles will require an integrated approach that blends physical models, geographical laws, and advanced DL techniques, while also considering the unique complexities of spatiotemporal data.

Given the significance of these challenges, coupled with the growing advancements in using DL to tackle them, there is a significant need for a comprehensive review of DL applications in spatiotemporal forecasting within ESS. This review references the rapidly evolving DL techniques and their applications in this field, aiming to provide a deeper understanding and utilization of the spatiotemporal dependence of Earth system phenomena and facilitate the integration of DL with domain-specific knowledge. Furthermore, this review identifies gaps in existing methodologies and suggests new research directions, ultimately aiding more Earth system scientists in adopting these advanced DL methods.

2. Challenges of spatiotemporal forecasting in ESS

Considering the pressing challenges and compelling opportunities outlined above, we proceed to delve into the multifaceted challenges of spatiotemporal forecasting within ESS (Section 2.1), as

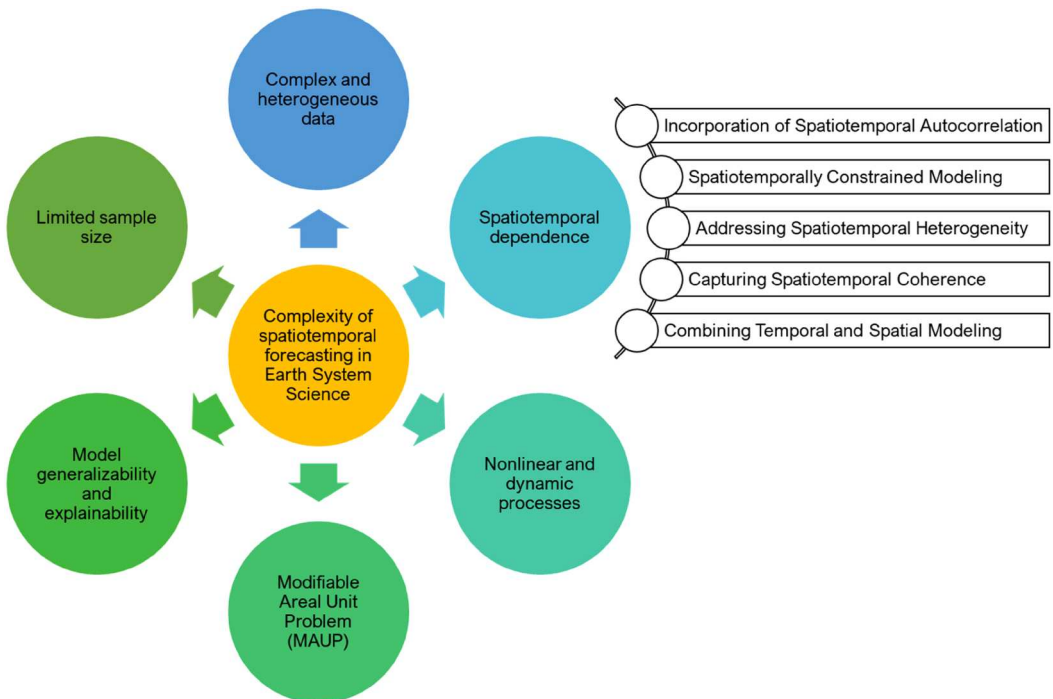


Figure 1. Schematic representation of the complex challenges associated with spatiotemporal forecasting in ESS (Section 2.1), and the main aspects of spatiotemporal dependence (Section 2.2).

illustrated in Figure 1. We then discuss the necessity of understanding spatiotemporal dependence and examine the techniques to address it in forecasting models (Section 2.2).

2.1. Complexities of spatiotemporal forecasting in ESS

Spatiotemporal forecasting in ESS is a challenging task due to the complexity and heterogeneity of the data involved (Hamdi et al. 2022). Satellite imagery, numerical simulations, and environmental monitoring are often dynamic and large in volume. These datasets come with diverse scales and formats, ranging from time-varying imagery to time-series measurements, and from structured numerical data to unstructured text, image, or video data (Scheele, Yu, and Huang 2021). Integrating multiple heterogeneous datasets with different scales, coverage, and quality makes the process even more complicated (Yu, Masrur, and Blaszczyk-Boxe 2023).

Spatiotemporal phenomena, such as climate or meteorological phenomena, are often nonlinear and dynamic processes, and the interconnections between influencing factors of these phenomena are also complex (Hashemi 2021). In addition, observations that directly capture the phenomena are typically limited due to the spatiotemporal complexity of the phenomena, constraints in data collection, and specific application requirements. For instance, monitoring wildfire spread in a large forest area may require ground measurements or satellite imagery covering the entire region. However, such observations are limited due to the limited availability of ground sensors or the frequency of satellite overpasses (Chen et al. 2022a).

Addressing model generalizability is crucial in the modeling process and has a significant impact on performance. An increasing number of DL models offer an improvement in generalizability by learning more robust features from high-dimensional data (Yuan et al. 2020), but they face challenges when trained on limited data. Such models may overfit, capturing noise instead of underlying patterns, thus affecting both their predictive performance and interpretability (Zang et al. 2021).

In addition, the way continuous spatial and temporal phenomena are discretized into regions or intervals can significantly impact the outcomes of DL models, as different choices of spatial and temporal resolutions can reveal or hide various patterns. The geographic scale of the study, which significantly influences model performance, and MAUP – a statistical bias resulting from aggregating point-based spatial data into larger units – can distort model outcomes if not properly addressed (Liu and Biljecki 2022).

2.2. Complexities of spatiotemporal dependence

Among the challenges in spatiotemporal forecasting in ESS (Section 2.1), the issue of spatiotemporal dependence is particularly complex. Therefore, this section will decouple this challenge in detail.

2.2.1. Incorporating spatiotemporal autocorrelation

Spatiotemporal autocorrelation refers to the statistical relationship or correlation between observations of a variable at different locations (spatial) and times (temporal). To incorporate spatiotemporal autocorrelation in the application of DL models to ESS, techniques like inverse distance weighting or kernel-based weighting can be leveraged to capture the inherent autocorrelation of environmental parameters by assigning weights based on the spatial proximity of observations (Li et al. 2017; Ma et al. 2019; Yousefzadeh, Hosseini, and Farnaghi 2021). These autocorrelations can then be used alongside other predictors as input variables to the DL models. However, since the autocorrelation features are treated as additional independent input features rather than inherently intertwined properties of the spatiotemporal data (Yuan et al. 2020), more sophisticated approaches, such as hierarchical models and Gaussian processes, can better exploit the structural dependencies inherent in spatiotemporal data (Wikle and Zammit-Mangion 2023). Examples can be interactions between physical and biological variables of the Earth system, e.g. how changes

in atmospheric composition can alter phenology, growing season length, and community composition of terrestrial ecosystems (Bonan and Doney 2018).

2.2.2. Considering spatiotemporal dependencies

Traditional methodologies often prioritize either the spatial or temporal dimension, neglecting the inherent interplay between the two. Holistic spatiotemporal models are essential to capture the complex behavior of Earth systems, where events propagate over time to influence nearby locations. DL models like convolutional LSTM (ConvLSTM) have been developed to capture both spatial and temporal aspects (Chen et al. 2020; Shi et al. 2015). Hybrid models that integrate DL with classical statistical models, such as nested spatial processes and deep compositional spatial models, offer a promising approach to capturing these dependencies (Zammit-Mangion et al. 2022). Random process models, including methods such as stochastic processes, Gaussian processes, and Markov random fields, are well-suited for modeling the probabilistic nature of environmental variables. For instance, hierarchical models and Gaussian processes can better exploit the structural dependencies inherent in spatiotemporal data by capturing the stochastic variability and uncertainty present in Earth system processes (Cressie and Wikle 2015; Wikle and Zammit-Mangion 2023). These perspectives emphasize the importance of incorporating randomness and uncertainty explicitly in the modeling process to enhance predictive performance and interpretability. Alternatively, regularization methods have been explored to enforce these constraints within the learning process, enabling a deep understanding of the spatial and temporal structures in the data (Cui et al. 2020). However, incorporating spatiotemporal aspects into a comprehensive spatiotemporal model introduces new challenges, including complex data structures requiring advanced data management, increased computational resources due to higher data dimensionality, potential anisotropy and non-stationarity requiring careful consideration, and the impact of spatiotemporal scales on model performance (Tascikaraoglu 2018).

2.2.3. Addressing spatiotemporal heterogeneity

Another key challenge relates to spatiotemporal heterogeneity, which is the variation of environmental variables across space and time. Geographically and Temporally Weighted DL (GTW-DL) models (Li et al. 2020) can capture local variations in predictor-target relationships. Instead of applying a global model that assumes stationarity in these relationships, GTW-DL establishes separate models for different locations and times, thereby addressing the spatiotemporal heterogeneity. However, challenges for implementing GTW-DL models include limited sample availability in a spatiotemporally localized context, leading to potentially less robust models and overfitting issues. In addition, as GTW-DL models create multiple local models instead of a single global one, the increased computational cost can be problematic, especially when dealing with extensive geographical areas and longtime spans. Recent advancements in hybrid models that integrate DL with statistical models, such as DeepKriging, have shown promise in capturing complex spatial and temporal dependencies without requiring explicit covariance functions or stationarity assumptions (Nag, Sun, and Reich 2023). However, specific challenges remain in optimizing these models to handle unevenly distributed and sparse data efficiently, reducing the computational load for large-scale applications, and improving the integration of diverse data sources to maintain model accuracy and robustness.

2.2.4. Capturing spatiotemporal coherence

Earth system phenomena can show strong systematic variation across scales – i.e. spatiotemporal coherence – with structures such as atmospheric blocking, tropical cyclones, and the El Niño–Southern Oscillation (Kashinath et al. 2021). Ensuring that DL models accurately capture this underlying spatiotemporal coherence is crucial, but conventional DL models may not be explicitly designed to capture the coherent structures, learn physically plausible structures, or robustly handle multi-scale variations. Hybrid models, such as those incorporating deep Gaussian processes (GPs)

and nested spatial processes, offer enhanced capabilities for capturing these coherent structures (Damianou and Lawrence 2013). Several studies demonstrate that infinitely wide deep neural networks (DNNs) converge to GPs, enabling Bayesian inference for deep networks (Lee et al. 2018). Researchers have developed methods to combine DNNs and GPs, such as using DNNs as feature mapping functions for GPs (Huang et al. 2015), approximating Bayesian DNNs with GP posteriors (Khan et al. 2019), and interpreting DNNs as point estimates for deep GPs (Dutordoir et al. 2021). However, significant challenges remain, including the high computational demands of training and inference, difficulties in scaling to large and high-dimensional datasets, dependency on high-quality and comprehensive data, complexity in integrating and interpreting model components, and ensuring robustness and generalization to unseen data.

3. Methodology

We retrieved the papers in the Scopus database by searching the keywords ‘deep learning’ and ‘spatiotemporal.’ The obtained 3669 papers were filtered to include journal papers only from January 2010 until April 2023, in the areas of engineering, environmental science, and Earth and planetary sciences. From this reduced list of 1591 papers, we applied two major refining criteria: (i) the papers should be based on the DL models, (ii) the applications must address Earth system processes, and (iii) the DL task is spatiotemporal forecasting. This procedure resulted in 56 reviewable papers. We then extended this list through a snowball search, which involved considering cited and citing works. This resulted in a representative subset of 72 eligible papers (Figure 2), which is considered a representative subset to provide a comprehensive review of recent advances and developments in this area.

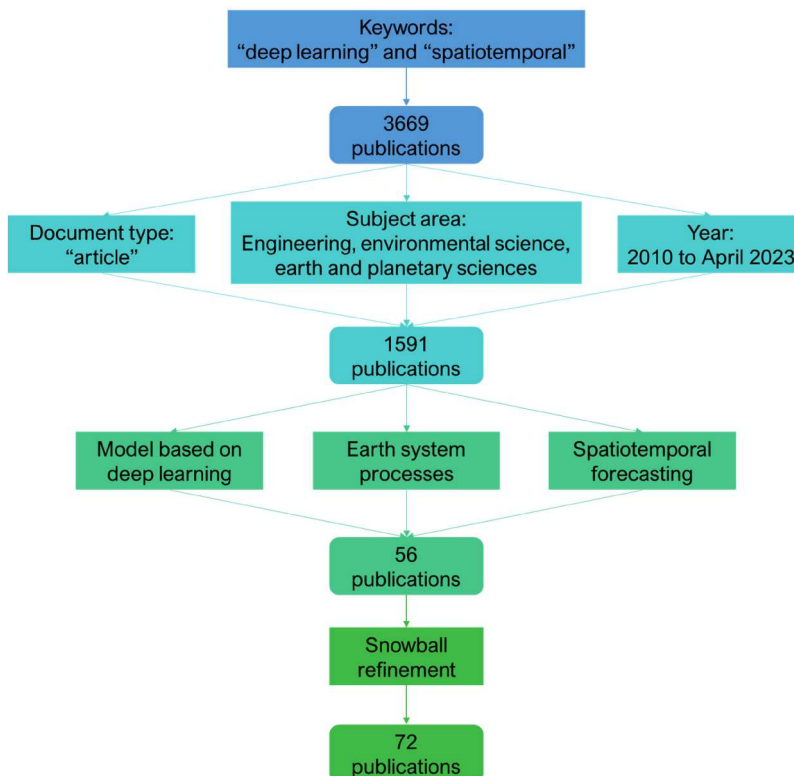


Figure 2. Flowchart of the methodology applied for the process of paper selection.

Table 1. DL applications for spatiotemporal forecasting in ESS. References are classified in terms of application field and subject of the study, nature of the prediction task, and the DL model used.

Application field: subject of the study	Model
Prediction task	
Agriculture: soil moisture, evapotranspiration	
Spatiotemporal forecasting: time-varying raster	ConvLSTM (Li et al. 2021)
Time series forecasting	LSTM (Talib et al. 2021)
Air quality prediction: Benzene, PM2.5, Ozone, NOx	
Spatiotemporal data fusion	Ensemble machine learning (Hsu et al. 2022; Lyu et al. 2019; Zang et al. 2021) Temporal: attention-based sequence-to-sequence model + Spatial: DNN (Wang et al. 2020) Spatio-temporal DeepKriging (Nag, Sun, and Reich 2023)
Spatiotemporal forecasting: time-varying raster	U-Net (He et al. 2022)
Time series forecasting	Attention-based sequence-to-sequence model (Jia, Cao, and Yang 2021; Mao et al. 2021; Xu et al. 2020) 3D CNN and gated recurrent unit (3D CNN-GRU) (Faraji et al. 2022) Dynamic graph convolutional network and the multi-channel temporal convolutional network (DGC-MTCN) (Dun, Yang, and Lei 2022) CNN-LSTM (Pak et al., 2020; Zhang et al. 2022) LSTM-based (Mao et al. 2021; Seng et al., 2021; Xu et al. 2020) Transformer-based (Yu, Masrur, and Blaszcak-Boxe 2023)
Climate modeling and weather prediction: Airstream, cloud, ENSO, land surface temperature, precipitation, sea ice concentration, sea surface height, sea surface temperature, tropical cyclone	
Spatiotemporal forecasting: time-varying raster	U-Net (Quinting et al. 2022) Temporal: ConvLSTM + Spatial: U-Net and convolutional block attention module (CBAM) (Wu et al. 2022) Spatiotemporal LSTM (Lian and Chen 2023; Xu et al. 2020) Autoencoder (Saha and Nanjundiah 2020) ConvLSTM (Hou et al. 2022; Kartal and Sekertekin 2022; Xiao et al. 2019; Zhou et al., 2021) ConvLSTM with spatial attention network (Dong et al. 2022) Spatiotemporal graph convolutional networks (Zhang et al. 2022) Pseudo flow spatiotemporal LSTM unit (PFST-LSTM) (Luo, Li, and Ye 2021) Attention-based (Che et al. 2022; Ren et al. 2023) Conditional GAN (Ravuri et al. 2021; Wang, Tang, and Gentine 2021)
Disaster management: earthquake, flood, lightning, wildfire, landslide	
Classification	DNN (Kong, Martin-Short, and Allen 2020; Yousefzadeh, Hosseini, and Farnaghi 2021) Spatiotemporal graph convolutional networks (Yuan et al. 2022)
Spatiotemporal forecasting: time-varying raster	LSTM and reduced order model (ROM) (Hu et al. 2019) Transformer (Lu et al. 2022) ConvLSTM (Kondylatos et al. 2022) Spatial: CNN + Temporal: LSTM (Zhang et al. 2022)
Time series forecasting	ConvLSTM (Chen et al. 2022a) Temporal graph convolutional network-LSTM (TGCN-LSTM) model and an Attention-TGCN model (Xi et al. 2023)
Environmental monitoring: water quality, algal bloom, NDVI, water quality	
Regression	DNN (Zheng et al. 2023)
Spatiotemporal forecasting: time-varying raster	ConvLSTM (Tang et al. 2022) DNN (Das and Ghosh 2016; Ramaraj and Sivakumar 2023) Deep Compositional Spatial Models (Zammit-Mangion et al. 2022)
Time series forecasting	Spatial: CNN + Temporal: BiLSTM (Ni et al. 2022)
Hydrological modeling and prediction: groundwater, runoff	
Regression	ANN and SVM (Malakar et al. 2023) DNN and CNN (Wang, Wang, and Han 2022)
Time series forecasting	ConvLSTM (Tyson et al. 2023) CNN-LSTM (Li et al. 2022) Spatiotemporal graph convolution (Deng et al. 2023) 3D-CNN – Transformer (Ghobadi and Kang 2022)

(Continued)

Table 1. Continued.

Application field: subject of the study	
Prediction task	Model
Oceanography: ocean eddy, ocean temperature, ocean wave, ocean current	
Spatiotemporal forecasting: time-varying raster	Autoencoder equipped with channel and spatial attention mechanisms (CSA-encoder) + GRU with temporal attention (Wang et al. 2022) Temporal: Conv1D + LSTM (Xiao et al. 2022) ConvLSTM (Li et al. 2022)
Time series forecasting	Spatial Channel Attention Module (SCAM) + Gated-Recurrent-Unit (GRU) (Xie et al. 2023) LSTM + DNN (Linghu et al. 2023)
Renewable energy forecasting: wind power	
Regression	3D-CNN (Zhu et al. 2021)
Spatiotemporal forecasting: time-varying raster	Autoencoder (Yu et al. 2019) DNN (Zhang and Zhao 2021)
Time series forecasting	Graph attention network + RNN (Chengqing et al. 2023; Liao et al. 2022; Ren et al. 2023; Wang, Wang, and Han 2022) Spatiotemporal dynamics attention (Ye et al. 2022)

The selected papers are categorized in Table 1, which illustrates the major details, including the application field and study object, the nature of the prediction task, and the DL model used. DL architectures that are commonly used for spatiotemporal forecasting in ESSs are summarized in Section 4. General findings related to technical innovations are presented in Section 5. Specific findings for each application field are then presented in Section 6. These specific sections provide an in-depth discussion of the DL models employed, with a focus on the architecture, contribution, and innovation.

4. DL methods for spatiotemporal forecasting

Here, we briefly summarize the DL architectures that are being used for spatiotemporal forecasting in ESS (Figure 3). The basic modules used in these architectures involve recurrent neural networks

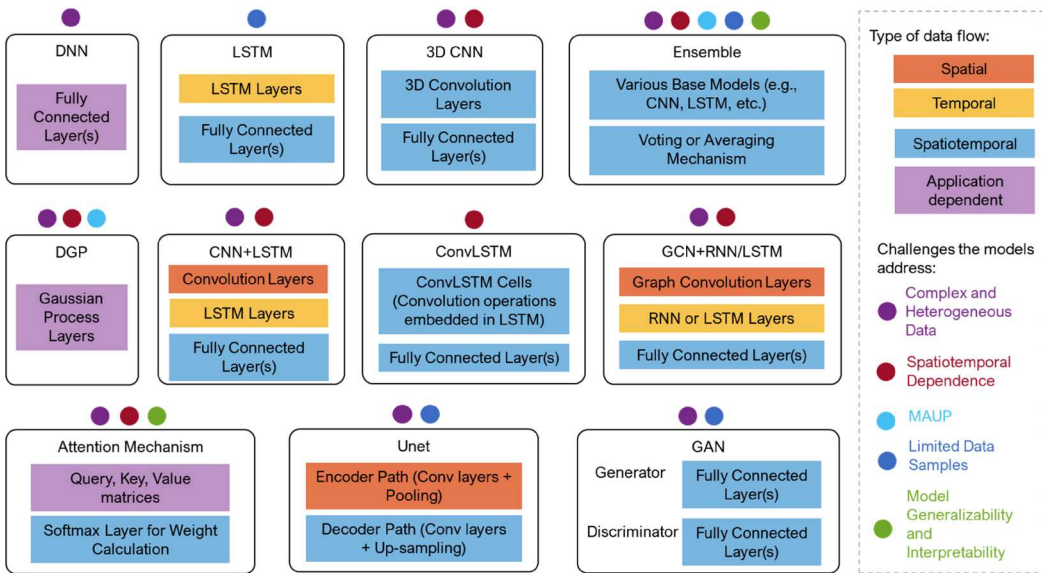


Figure 3. Comparative architecture and capabilities of various DL models for spatiotemporal forecasting. This figure visually represents the architecture of commonly used DL models – 3D CNN, Attention Mechanism, CNN + LSTM, ConvLSTM, DNN, Ensemble, GAN, GCN + RNN, LSTM, and U-Net – and highlights their ability to address specific challenges in spatiotemporal forecasting. All the models here can address the challenge of handling nonlinear and dynamic processes.

Table 2. Base models for spatiotemporal forecasting in ESS.

Base model		References	Count of Reference
Application field			
3D CNN			2
Air quality prediction	Faraji et al. (2022)		1
Renewable energy forecasting	Zhu et al. (2021)		1
Attention			14
Air quality prediction	Wang et al. (2020); Jia, Cao, and Yang (2021); Zhu et al. (2021); Yu, Masrur, and Blaszcak-Boxe (2023)		4
Climate modeling and weather prediction	Che et al. (2022); Ren et al. (2023)		2
Disaster management	Lu et al. (2022)		1
Hydrological modeling and prediction	Ghobadi and Kang (2022)		1
Oceanography	Wang et al. (2022); Xie et al. (2023)		2
Renewable energy forecasting	Chengqing et al. (2023); Wang and He (2022); Ren et al. (2023); Ye et al. (2022)		4
CNN + LSTM			6
Air quality prediction	Pak et al. (2020)		1
Climate modeling and weather prediction	Xu et al. (2020)		1
Disaster management	Zhang et al. (2022)		1
Environmental monitoring	Ni et al. (2022)		1
Hydrological modeling and prediction	Li et al. (2022)		1
Oceanography	Xiao et al. (2022)		1
ConvLSTM			14
Agriculture	Li et al. (2021)		1
Air quality prediction	Zhang, Wang, and Liu (2022)		1
Climate modeling and weather prediction	Wu et al. (2022); Lian and Chen (2023); Kartal and Sekertekin (2022); Zhou et al. (2021); Xiao et al. (2019); Hou et al. (2022); Dong et al. (2022)		7
Disaster management	Kondylatos et al. (2022); Chen et al. (2022)		2
Environmental monitoring	Tang et al. (2022)		1
Hydrological modeling and prediction	Tyson et al. (2023)		1
Oceanography	Li et al. (2022)		1
DNN			8
Disaster management	Kong, Martin-Short, and Allen (2020); Yousefzadeh, Hosseini, and Farnaghi (2021)		2
Environmental monitoring	Zheng et al. (2023); Das and Ghosh (2016); Ramaraj and Sivakumar (2023)		3
Hydrological modeling and prediction	Malakar et al. (2023); Wang, Wang, and Han (2022)		2
Renewable energy forecasting	Zhang and Zhao (2021)		1
Ensemble			3
Air quality prediction	Hsu et al. (2022); Lyu et al. (2019); Zang et al. (2021)		3
GAN			2
Climate modeling and weather prediction	Ravuri et al. (2021); Wang, Tang, and Gentine (2021)		2
GCN + RNN			8
Air quality prediction	Mao et al. (2021); Wu et al. (2023); Dun, Yang, and Lei (2022)		3
Climate modeling and weather prediction	Zhang et al. (2022)		1
Disaster management	Yuan et al. (2022); Xi et al. (2023)		2
Hydrological modeling and prediction	Deng et al. (2023)		1
Renewable energy forecasting	Liao et al. (2022)		1
LSTM			8
Agriculture	Talib et al. (2021)		1
Air quality prediction	Xu et al. (2020); Li et al. (2017); Mao et al. (2021); Seng et al. (2021)		4
Climate modeling and weather prediction	Luo, Li, and Ye (2021)		1
Disaster management	Hu et al. (2019)		1

(Continued)

Table 2. Continued.

Base model		Count of
Application field	References	Reference
Oceanography	Linghu et al. (2023)	1
Unet		2
Air quality prediction	He et al. (2022)	1
Climate modeling and weather prediction	Quinting et al. (2022)	1
Statistical Deep Learning		3
Satellite imagery gap-filling	Zammit-Mangion et al. (2022)	1
Air quality prediction	Nag, Sun, and Reich (2023)	1
Review paper	Wikle and Zammit-Mangion (2023)	1

(RNNs) for temporal modeling, convolutional neural networks (CNNs) for spatial modeling, graph neural networks (GNNs) for spatiotemporal data with underlying graph-like structure, generative adversarial networks (GANs) for generating synthetic spatiotemporal data, attention models (such as Transformer) that focus on important parts of the input data and have shown effectiveness in both spatial and temporal modeling (Table 2). Many advanced DL architectures used in spatiotemporal forecasting are hybrids or combinations of the base models, created to leverage the strengths of different models and address the unique challenges of spatiotemporal data.

4.1. ConvLSTM

ConvLSTM is a neural network architecture that combines the spatial modeling capabilities of CNNs with the temporal modeling abilities of long short-term memory (LSTM) networks (Shi et al. 2015). ConvLSTM introduces convolutional operations to both the input-to-state and state-to-state transitions of the LSTM, allowing the model to incorporate spatial and temporal dependencies. ConvLSTM is formulated as follows:

Input Gate: The input gate decides how much new information will be stored in the cell state. It uses the current input $X_t \in R^{H \times W \times C}$ (where H is the height, W is the width, and C is the number of channels of the input feature map), the previous hidden state $H_{t-1} \in R^{H \times W \times D}$ (where D is the number of channels in the hidden state), and the previous cell state $C_{t-1} \in R^{H \times W \times D}$ to compute the input gate i_t with a sigmoid function, which converts the values between 0 and 1.

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \odot C_{t-1} + b_i) \quad (1)$$

where $W_{xi} \in R^{K \times K \times C \times D}$ is the weight matrix for the input gate applied to the input X_t ; $W_{hi} \in R^{K \times K \times D \times D}$ is the weight matrix for the input gate applied to the hidden state H_{t-1} ; $W_{ci} \in R^D$ is the weight matrix for the input gate applied to the cell state C_{t-1} ; $b_i \in R^D$ is the bias for the input gate; $i_t \in R^{H \times W \times D}$ is the input gate output; σ is the sigmoid function; $*$ is the convolutional operation with kernel size $K \times K$; $\odot$ is the element-wise multiplication.

Forget Gate: The forget gate decides how much the previous cell state to retain. It is computed using the current input, the previous hidden state, and the previous cell state through a sigmoid function.

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \odot C_{t-1} + b_f) \quad (2)$$

where $W_{xf} \in R^{K \times K \times C \times D}$ is the weight matrix for the forget gate applied to the input X_t ; $W_{hf} \in R^{K \times K \times D \times D}$ is the weight matrix for the forget gate applied to the hidden state H_{t-1} ; $W_{cf} \in R^D$ is the weight matrix for the forget gate applied to the cell state C_{t-1} ; $b_f \in R^D$ is the bias for the forget gate; and $f_t \in R^{H \times W \times D}$ is the forget gate output.

New Memory Cell: The new memory cell represents the new information to store in the cell state and is computed using the current input and the previous hidden state.

$$\tilde{C}_t = \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (3)$$

where $W_{xc} \in R^{K \times K \times C \times D}$ is the weight matrix for the new memory cell applied to the input X_t ; $W_{hc} \in R^{K \times K \times D \times D}$ is the weight matrix for the new memory cell applied to the hidden state H_{t-1} ; $b_c \in R^D$ is the bias for the new memory cell; and $\tilde{C}_t \in R^{H \times W \times D}$ is the new memory cell output.

Final Memory Cell: The final memory cell is updated by retaining certain information from the previous cell state (controlled by the forget gate) and adding some new information (controlled by the input gate and the new memory cell).

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (4)$$

where $C_t \in R^{H \times W \times D}$ is the final memory cell state.

Output Gate: The output gate decides what part of the cell state will be outputted. It uses the current input, the previous hidden state, and the updated cell state to compute the output gate using a sigmoid function.

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \odot C_t + b_o) \quad (5)$$

where $W_{xo} \in R^{K \times K \times C \times D}$ is the weight matrix for the output gate applied to the input X_t ; $W_{ho} \in R^{K \times K \times D \times D}$ is the weight matrix for the output gate applied to the hidden state H_{t-1} ; $W_{co} \in R^D$ is the weight matrix for the output gate applied to the cell state C_t ; $b_o \in R^D$ is the bias for the output gate; and $o_t \in R^{H \times W \times D}$ is the output gate output.

Hidden State: The hidden state is updated by applying the tanh function to the cell state and multiplying it by the output gate. This gives the final output of the ConvLSTM at the current time step.

$$H_t = o_t \odot \tanh(C_t) \quad (6)$$

where $H_t \in R^{H \times W \times D}$ is the hidden state at time t .

4.2. Graph Convolutional Networks (GCNs) and Recurrent Neural Networks (RNNs)

This combination allows for capturing both spatial relationships (via the GCN) and temporal dynamics (via the RNN). This is particularly useful in scenarios where data is represented as a graph and evolves over time, such as in traffic forecasting or social network analysis (Feng et al. 2022; Yu, Yin, and Zhu 2018).

A GCN layer can be formatted as follows:

$$H_{l+1} = \sigma(D^{-\frac{1}{2}} A D^{-\frac{1}{2}} H_l W_l) \quad (7)$$

where $H_l \in R^{N \times F_l}$ is the matrix of node features at the l -th layer (H^0 is the input feature matrix); $A \in R^{N \times N}$ is the adjacency matrix of the graph; $D \in R^{N \times N}$ is the degree matrix of the graph; $W_l \in R^{F_l \times F_{l+1}}$ is the weight matrix of the l -th layer; σ is a non-linear activation function; N is the number of nodes; F_l and F_{l+1} are the number of features in the l -th and $(l+1)$ -th layers, respectively. The operation $D^{-\frac{1}{2}} A D^{-\frac{1}{2}} H^l$ is the graph convolution operation, which applies a localized first-order approximation of spectral graph convolutions.

A RNN layer can be defined as follows:

$$h_t = \sigma(W_{hh} h_{t-1} + W_{xh} x_t + b_h) \quad (8)$$

$$y_t = W_{hy} h_t + b_y \quad (9)$$

where $h_t \in R^H$ is the hidden state at time t ; $h_{t-1} \in R^H$ is the hidden state at the previous time step; $x_t \in R^F$ is the input at time t ; $y_t \in R^O$ is the output at time t ; $W_{hh} \in R^{H \times H}$, $W_{xh} \in R^{F \times H}$, and $W_{hy} \in R^{H \times O}$ are the weight matrices; $b_h \in R^H$ and $b_y \in R^O$ are the bias terms; σ is a non-linear activation function; H is the number of hidden units; F is the number of input features; and O is the number of output features.

In models that combine GCNs and RNNs, the output of the GCN (H^L), where L is the number of GCN layers) would be used as the input to the RNN (x_t). This allows the model to capture both the spatial structure of the data (through the GCN) and the temporal dynamics (through the RNN).

4.3. 3D-CNN

3D-CNN is a CNN architecture that can handle spatiotemporal data with three dimensions, such as video data or volumetric medical images. 3D-CNN extends the traditional 2D convolution operation to 3D by adding a temporal dimension, allowing the network to learn spatial and temporal features simultaneously (Zhu et al. 2021).

A 2D convolution operation in a 2D CNN for a given filter can be mathematically defined as:

$$f(i, j) = \sum_m \sum_n \text{input}(i - m, j - n) \cdot \text{filter}(m, n) \quad (10)$$

Where $\text{input} \in R^{H \times W}$ is the 2D input matrix, such as an image, where H is the height and W is the width; $\text{filter} \in R^{k_H \times k_W}$ is the 2D filter matrix with height k_H and width k_W ; and $f(i, j)$ is the convolution result at position (i, j) .

In contrast, a 3D convolution operation in a 3D CNN for a given filter is defined as:

$$f(i, j, k) = \sum_m \sum_n \sum_p \text{input}(i - m, j - n, k - p) \cdot \text{filter}(m, n, p) \quad (11)$$

where $\text{input} \in R^{H \times W \times D}$ is the 3D input tensor, where H is the height, W is the width, and D is the depth (temporal dimension); $\text{filter} \in R^{k_H \times k_W \times k_D}$ is the 3D filter tensor with height k_H , width k_W , and depth k_D ; and $f(i, j, k)$ is the convolution result at position (i, j, k) .

4.4. UNet

UNet is a CNN architecture widely used for image segmentation tasks but can also be adapted for spatiotemporal data analysis (Bauer et al., 2015; Fu et al., 2022). UNet uses contracting and expanding paths to capture global and local information in the input data. The contracting path consists of convolutional and max pooling layers that capture the spatial features of the data, while the expanding path uses transposed convolutional layers to reconstruct the output.

UNet models that are extended to 3D spatiotemporal data have usually adjusted the contracting (downsampling) and expanding (upsampling) paths of the U-Net to use 3D convolutions, similar to adjusting a traditional CNN to a 3D CNN.

On the contracting path, for each layer l from 1 to L , the operations can be formulated as:

$$\text{3D Convolution: } X_l = \text{Conv3D}(X_{l-1}, F_l, \text{stride} = 1) \quad (12)$$

$$\text{Activation: } X_l = \text{ReLU}(X_l) \quad (13)$$

$$\text{Max Pooling: } X_l = \text{MaxPool3D}(X_l, \text{pool-size} = 2, \text{stride} = 2) \quad (14)$$

where $X_{l-1} \in R^{H_{l-1} \times W_{l-1} \times D_{l-1} \times C_{l-1}}$ is the output from the previous layer, where H_{l-1} is the height, W_{l-1} is the width, and D_{l-1} is the depth (temporal dimension), and C_{l-1} is the number of channels. $F_l \in R^{k_H \times k_W \times k_D \times C_{l-1} \times C_l}$ is the 3D filter for layer l , where k_H , k_W , and k_D are the dimensions of the filter (height, width, and depth), and C_l is the number of output channels. Conv3D denotes the 3D convolution operation. X_l is the output tensor of the current layer. ReLU is the Rectified Linear Unit activation function applied element-wise. MaxPool3D reduces the spatial dimensions by a factor of 2.

On the expanding path, for each layer l from $L + 1$ to $2L$, the operations can be formulated as:

$$\text{Upsampling: } X_l = \text{Upsample}(X_{l-1}) \quad (15)$$

$$\text{Concatenation: } X_l = \text{Concat}(X_l, X_{2L-l}) \quad (16)$$

$$\text{3D Convolution: } X_l = \text{Conv3D}(X_l, F_l, \text{stride} = 1) \quad (17)$$

where Upsample doubles the size of the input along each dimension, Concat concatenates the upsampled output with the output from the corresponding layer in the contracting path (X_{2L-l}). And the 3D Convolution is followed by an activation operation as Eq. 13.

After the last layer in the expanding path, a final 3D convolution to get the output can be formulated as:

$$Y = \text{Conv3D}(X_{2L}, F_{out}, \text{stride} = 1) \quad (18)$$

where $F_{out} \in R^{k_H \times k_W \times k_D \times C_{2L} \times C_{out}}$ is the 3D filter for the output layer; $Y \in R^{H_0 \times W_0 \times D_0 \times C_{out}}$ is the final output, where H_0 is the height, W_0 is the width, D_0 is the depth, and C_{out} is the number of output channels.

4.5. Attention-based models

Attention-based models use attention mechanisms, such as self-attention, channel attention, or spatial attention, to attend to relevant features in the spatiotemporal data (Chengqing et al. 2023; Wang and He 2022; Zhu et al. 2021). For example, a channel attention mechanism can focus on certain channels of the input data that are more relevant for a predicting task, while a spatial attention mechanism can learn to attend to specific geographic regions of the input data. Temporal attention mechanisms can learn to focus on specific time steps in a sequence. There are multiple strategies to combine these attentions, such as a sequential combination of spatial and temporal attentions, or a 3D attention mechanism across spatial and temporal dimensions.

As a simplified version, the latter can be formulated with an input tensor X of shape (T, H, W, C) , where T is the number of time steps, H is the height, W is the width, and C is the number of channels. First, an importance score for each grid in the input tensor can be calculated, such as by applying a suitable function (e.g. a $1 \times 1 \times 1$ convolution or a multi-layer perceptron) to the input tensor X to obtain a new tensor A of shape $(T, H, W, 1)$:

$$A = F(X) \quad (19)$$

where $A \in R^{T \times H \times W \times 1}$ is the tensor of importance scores, F is a function that transforms the input tensor X , and $X \in R^{T \times H \times W \times C}$ is the input tensor, where T is the number of time steps, H is the height, W is the width, and C is the number of channels.

A softmax function over the entire tensor A can be applied to compute the spatiotemporal attention weights for each grid:

$$W_{t,i,j} = \frac{\exp(A_{t,i,j})}{\sum_p \sum_q \sum_r \exp(A_{p,q,r})} \quad (20)$$

where $W_{t,i,j} \in R^{T \times H \times W}$ represents the attention weights, $A_{t,i,j}$ is the element of the tensor A at time t , height i , and width j . The indices p, q, r run over all time steps, heights, and widths, respectively, used in the softmax normalization to ensure that the weights sum to 1 across all spatiotemporal locations.

The input tensor X is finally multiplied by the attention weights W to obtain the output tensor Y :

$$Y = W \odot X \quad (21)$$

where $Y \in R^{T \times H \times W \times C}$ is the output tensor, and the $\odot$ operation denotes element-wise multiplication, applied independently for each channel. This means that the feature map at each spatiotemporal location in the input tensor X is multiplied by its corresponding attention weight in W .

4.6. GAN-based models

Generative Adversarial Networks (GANs) have been adapted to handle spatiotemporal forecasting. GANs consist of two components: a generator network G and a discriminator network D . The generator creates new samples, and the discriminator tries to distinguish between real and generated samples. When adapting GANs to handle spatiotemporal data, and specifically for tasks like time-varying imagery forecasting, Conditional GANs (cGANs) incorporate spatial and temporal contexts directly into their architecture, enabling the generation of new data samples that respect these contexts for future predictions, such as precipitation nowcasting (Ravuri et al. 2021).

In a cGAN, the generator and the discriminator both condition on extra information c , such as historical precipitation observations. Assuming z is a noise vector drawn from a prior distribution, typically a multivariate normal distribution. The generator G produces a sample conditioned on z and c :

$$G(z, c) = X' \quad (22)$$

where $G(z, c) \in R^{T \times H \times W \times C}$ is the generated sample X' , $z \in R^d$ is the noise vector, and $c \in R^{T \times H \times W \times C}$, with T representing the number of time steps, H the height, W the width, C the number of channels, and d the number of hidden dimensions.

The discriminator D takes as input a sample X (either a real sample or a generated sample X') and the condition c , and outputs a probability that the sample is real:

$$D(X, c) = p(\text{real}) \quad (23)$$

where $D(X, c)$ is the probability that the sample $X \in R^{T \times H \times W \times C}$ is real given the condition $c \in R^{T \times H \times W \times C}$.

The training process for a cGAN involves a two-player minimax process, where the generator tries to fool the discriminator by producing realistic samples, and the discriminator tries to correctly classify real and generated samples. This can be expressed by the following loss function, which is typically used for GANs:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data}(x)} [\log D(x|c)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z|c)| c))] \quad (24)$$

where $\mathbb{E}$ denotes the expectation, $x \sim p_{data}(x)$ represents real samples x drawn from the data distribution p_{data} , $z \sim p_z(z)$ represents noise vectors z drawn from the noise distribution p_z , c is the conditional information, $D(x|c)$ is the discriminator's output for a real sample x given the condition c , and $D(G(z|c)| c)$ is the discriminator's output for a generated sample $G(z|c)$ given the condition c . The first term in the loss function corresponds to the discriminator's ability to correctly classify real samples, and the second term corresponds to its ability to correctly classify generated samples. This architecture allows the GAN to generate new samples that are consistent with the spatiotemporal context c , which makes it suitable for tasks like time-varying imagery forecasting.

4.7. Statistical deep learning

Statistical models integrated with deep learning, such as deep Gaussian processes, deep ensembles, and variational inference in deep learning, offer powerful tools for spatiotemporal forecasting by

combining the strengths of both approaches (Damianou and Lawrence 2013; Wikle and Zammit-Mangion 2023). These models leverage the flexibility and expressiveness of deep learning while incorporating the probabilistic foundations and interpretability of statistical methods. For example, deep Gaussian processes (DGPs) are an extension of Gaussian processes (GPs) that stack multiple layers of GPs to model complex data structures across space and time while providing uncertainty quantification through inherent GP measures. This makes them particularly suitable for capturing the intricate spatiotemporal dependencies inherent in Earth system processes. The first layer of DGPs can be generally represented as:

$$h_1(x) \sim GP(\mu_1(x), k_1(x, x')) \quad (25)$$

where $x \in R^d$ is the input data vector, $h_1(x) \in R^{d_1}$ is the output of the first GP layer, $\mu_1(x) \in R^{d_1}$ is the mean function of the first GP layer, and $k_1(x, x') \in R^{d_1 \times d_1}$ is the covariance function. Here, d is the input dimension, and d_1 is the output dimension of the first GP layer.

Each subsequent layer takes the output of the previous layer as its input:

$$h_{i+1}(h_i) \sim GP(\mu_{i+1}(h_i), k_{i+1}(h_i, h'_i)) \quad (26)$$

where $h_i \in R^{d_i}$ is the output from the i^{th} GP layer, $h_{i+1}(h_i) \in R^{d_{i+1}}$, $\mu_{i+1}(h_i) \in R^{d_{i+1}}$, and $k_{i+1}(h_i, h'_i) \in R^{d_{i+1} \times d_{i+1}}$ are the output, the mean function, and the covariance function of the $i + 1^{th}$ GP layer, respectively. Here, d_i and d_{i+1} are the dimensions of the output of the i^{th} and $i + 1^{th}$ GP layers, respectively.

The final layer provides the model output:

$$y(h_{L-1}) \sim GP(\mu_L(h_{L-1}), k_L(h_{L-1}, h'_{L-1})) \quad (27)$$

where $y \in R^{d_y}$ is the final output, L is the number of layers, $h_{L-1} \in R^{d_{L-1}}$ is the output of the $L - 1^{th}$ GP layer, $\mu_L \in R^{d_y}$ and $k_L \in R^{d_y \times d_y}$ are the mean function and covariance function of the final GP layer. Here, d_y is the output dimension. DGPs enable the capture of complex non-linear dependencies and provide robust uncertainty quantification, making them particularly valuable in environmental and Earth system sciences, where such characteristics are essential.

Additionally, statistical deep learning encompasses other advanced methods, such as deep kernel learning (Wilson et al. 2015) and Bayesian neural networks (Wilson and Izmailov 2022), which also aim to enhance uncertainty quantification and model robustness. While DGPs are highlighted here, these additional techniques further demonstrate the versatility of statistical deep learning in handling spatiotemporal forecasting challenges.

5. Technical innovations in using DL for spatiotemporal forecasting

Based on our review, there are four major technical innovations of DL models in spatiotemporal forecasting. These four points include (1) physics-informed models that incorporate domain-specific knowledge to enhance their forecasting capabilities; (2) explainable AI, which aims to make the complex DL models more understandable and interpretable; (3) novel modifications of existing DL architectures, highlighting creative adaptations that improve performance or address specific challenges; and (4) incorporating uncertainties, discussing methods that quantify and integrate uncertainties to provide more reliable forecasts. Table 3 provides an overview of which technical innovations are most relevant for each of the key challenges in spatiotemporal forecasting.

5.1. Physics-informed models

Innovative strategies have combined physics-based understanding with DL techniques in predicting environmental phenomena. One common approach is indirectly incorporating domain knowledge

Table 3. Matrix showing the relationship between key challenges in spatiotemporal forecasting and the technical innovations designed to address them. A checkmark (✓) indicates that the respective technical innovation can potentially mitigate the corresponding challenge.

Challenges	Physics-informed Models	Explainable AI	Novel Modifications of DL	Incorporating Uncertainties
Spatiotemporal dependence	✓		✓	✓
Spatiotemporal Autocorrelation	✓		✓	✓
Spatiotemporally Constrained Modeling	✓			✓
Addressing Spatiotemporal Heterogeneity			✓	
Capturing Spatiotemporal Coherence			✓	✓
Model Generalizability	✓	✓	✓	
Model Interpretability		✓		
MAUP (Modifiable Areal Unit Problem)	✓			
Limited Sample Size				✓

into a DL model through historical datasets. For example, Xiao et al. (2019) used a reanalysis dataset of sea surface temperature (SST) to reflect the physical processes that influence SST. A ConvLSTM model used in this study implicitly learns about these physical influences as it learns to predict future SST based on past reanalysis. Similarly, when predicting SST, Hou et al. (2022) integrated domain knowledge into a DL model by considering short-wave radiation and long-wave radiation as external factors affecting SST.

Another common approach is to incorporate physical understanding using simulation outputs from physics-based numerical models. For instance, Tang et al. (2022) incorporated physical understanding by generating a hybrid dataset that combines satellite remote sensing images with a physics-based hydrodynamic-biological model simulating chlorophyll levels in the lake. This allowed the DL model to be informed by key ecological and environmental dynamics that are central to the formation of harmful algal blooms. Thus, the developed ConvLSTM model is grounded in the physical processes underlying harmful algal blooms.

In addition to using simulation outputs to generate a fused dataset, endeavors have been made to correct forecast biases generated from simulation models. Lyu et al. (2019) used simulation output from a deterministic numerical model, specifically the community multiscale air quality (CMAQ) model, to create a fused air quality dataset. Such simulation models might be biased due to inaccuracies in input data (emission rates, meteorological fields) and incomplete or approximate representations of the actual atmospheric processes. An ensemble DL model was developed to combine model simulations with ground-level monitoring measurements to provide more accurate spatiotemporal estimates of pollutant concentrations. Quinting et al. (2022) corrected forecast errors of warm conveyor belts (WCBs) from the European Centre for Medium-Range Weather Forecasts ensemble model and the ICOsahedral Nonhydrostatic model for understanding the complex dynamics of WCBs and their impacts on extratropical cyclones and anticyclonic potential vorticity anomalies. CNN models were developed to augment and improve upon the existing physical simulation models, and the approach is predicated on an understanding of the physical phenomena of extratropical cyclones and WCBs. Similarly, Saha and Nanjundiah (2020) proposed a DL approach to address the limitations of the National Centers for Environmental Prediction coupled forecast system model in failing to properly capture the relationships among the Equatorial Indian Ocean Oscillation (EQUINOO), the El Niño Southern Oscillation (ENSO), and inter-annual variability of the Indian monsoon. An autoencoder model was developed for unsupervised feature learning from climatic variables and to identify potential predictors, and ensemble machine learning models were then used to predict the climate indices: ENSO and EQUINOO.

As a more physics-constrained approach, Zhang and Zhao (2021) developed a framework of physics-informed neural networks (PINNs), where the physical laws, in the form of partial

differential equations, are included in the training of the model by encoding them into the loss function. The Navier-Stokes (NS) equations, which govern fluid dynamics and thus are very important for wind field simulations, are directly encoded into the training process of the deep neural network. This direct inclusion of the NS equations in the learning process allows the model to capture the complex, nonlinear dynamics of the incoming wind without simplifying the model complexity. The network is trained to minimize both the functional loss, based on the residues of the NS equations, and the measurement loss, derived from the comparison of the model's outputs and actual LIDAR observations. The functional loss represents the adherence of the model predictions to the physical laws of fluid dynamics, while the measurement loss represents the adherence of the model predictions to the real-world data.

5.2. Explainable AI

Increasing attention is being directed towards explainable AI (xAI) to enhance model transparency and understandability. Various methodologies have been developed to achieve this, evidenced by several recent studies. For example, Zang et al. (2021) leveraged decision trees, which are self-explainable, in their tree-based ensemble DL model for O_3 prediction. The model's interpretability allows insight into the spatiotemporal aspects of O_3 pollution and reveals the notable influence of specific districts and months on O_3 levels.

Attention-based models can provide some level of interpretability, allowing the model to identify which parts of the input data are more relevant to the output. By visualizing the attention weights assigned to each input feature or sequence element, one can gain insight into the model's decision-making process and identify the most important features or elements for the prediction. Zhu et al. (2021) proposed the Attention-based Parallel Network (APNet) that assigns weights to different feature states at varying times. The model extracts short-term and long-term temporal features simultaneously using an attention-based CNN-LSTM multilayer structure, which helps to capture the complex nonlinearity of $PM_{2.5}$ concentration and simulate the temporal and spatial dependence of $PM_{2.5}$ concentration simultaneously. Similarly, Yu, Masrur, and Blaszcak-Boxe (2023) developed the SpatioTemporal (ST)-Transformer that improved spatiotemporal predictions of $PM_{2.5}$ concentrations in wildfire-prone areas by incorporating a sparse attention mechanism.

SHapley Additive exPlanations (SHAP; Lundberg and Lee 2017) have also been used for DL model interpretation. SHAP explains models by connecting optimal credit allocation with local explanations using Shapley values from cooperative game theory (Shapley 1953). Shapley values provide a method for fairly distributing the total gains of a coalition among its players based on their contributions. In SHAP, each feature is considered as a player in a game where the 'gain' is the prediction from a model. The Shapley value of each feature indicates its contribution to the difference between the actual and average predictions for all instances. This allows us to interpret the model's output regarding its input features. To address the lack of physical mechanics in water quality prediction, Zheng et al. (2023) utilized the SHAP method to identify the significant variables and reason their influencing directions on the variation of each water quality parameter. Kondylatos et al. (2022) used SHAP values to diagnose model attributions by identifying the substantial contribution of wetness-related variables and unveiling the temporal focus of the models. They also observed the variability of the contribution of input variables across wildfire events, which hints at different wildfire mechanisms.

Another xAI strategy is to use transfer learning for explainability. Li et al. (2021) discussed the predictive ability of different factors, such as lagged soil temperature, soil temperature, season, and precipitation, in predicting soil moisture in the target domain. The paper proposed applying cross-source transfer learning with DL models for soil temperature prediction in newly built datasets, which can enhance the model's interpretability by leveraging pre-trained models in the source domain (ERA5-land) and fine-tuning them in the target domain.

5.3. Novel modifications of existing DL architectures

Many studies in the review propose novel modifications to existing DL architectures to address specific challenges in their respective domains. These modifications aim to improve prediction accuracy, incorporate spatiotemporal correlations, and enhance the modeling of spatial and temporal dependencies. The common theme among most of these studies is the utilization of RNN/LSTM, and attention mechanisms, integrated with convolutions or graph neural networks, to capture spatiotemporal dependencies.

GCNs: Models that leverage GCNs learn weights to propagate information between nodes in a graph, allowing them to capture complex relationships and interactions. The first approach primarily utilizes static graphs to capture spatial patterns and dependencies. Liao et al. (2022) proposed a graph neural network-based short-term power forecasting approach combining GCN and LSTM to capture renewable energy spatiotemporal correlations. The GCN is used to learn the spatial correlation between adjacent renewable energy monitors, while the LSTM learns the dynamic changes in power generation curves, allowing the model to capture spatiotemporal correlations of renewable energy monitors. Yuan et al. (2022) developed three spatiotemporal graph convolutional network (STGCN) models to predict the near-future flooding status of road segments by incorporating not only the status of the road segments but also the status of neighboring road segments using a DL framework on fine-grained traffic data. The models consider adjacency, distance, and elevation differences between road segments to predict road network flooding status during flood events.

The second approach captures the predefined spatiotemporal correlation between nodes in the graph. Zhang, Liu, and Wang (2022) employed adaptive attributed spatiotemporal affinities between spatiotemporal nodes, including spatiotemporal proximity and multi-dimensional meteorological attribute similarity using graph embedding. These affinities are used to construct a spatiotemporal affinity matrix. GCNs are then applied to make non-linear predictions based on the learned spatiotemporal affinity matrices. Ren et al. (2023) proposed a Data-based Adaptive Graph Learning Network (DAGLN) for multi-node offshore wind power forecasting. The paper designed a graph learning module that constructs predefined graphs using the dynamic time warping algorithm, capturing prior knowledge of the model. An adaptive graph learning algorithm was also introduced to capture spatial correlations hidden in wind speed time series data. The fused spatial graph compensated for the limitations of predefined topological structures, enabling a deeper analysis of the individuality of the predicted area. The paper integrated the GL module based on data with a matrix decomposition module, creating an end-to-end learning framework that maximizes the potential of wind speed sequence data in spatiotemporal modeling.

The third approach explicitly utilizes dynamic adjacency matrices to represent time-varying spatial correlations. Wu et al. (2023) combined three different types of models, residual neural network (ResNet), GCN, and bi-directional LSTM (BiLSTM), in a hybrid architecture to predict short-term regional NO₂ and O₃ concentrations. The Res-GCN-BiLSTM model leverages spatiotemporal correlations and topological relationships among air quality monitoring networks and auxiliary pollutants and meteorology to improve prediction accuracy. The authors also used auto-correlation analysis and cluster analysis to reveal the inherent temporal and spatial properties of monitoring networks, which were then adaptively integrated into the model. Dun, Yang, and Lei (2022) proposed a new correlation calculation method, called dynamic distance gray relation analysis (DDGRA), to construct dynamic adjacency matrices, capturing time-varying spatial correlations. They then used a hybrid DL architecture, DGC-MTCN (Dynamic Graph Convolutional-Multi-Channel Temporal Convolutional Network), that combines two powerful DL components, dynamic GCN and multi-channel temporal convolutional network (TCN). A GCN is utilized to extract spatial information from the dynamic adjacency matrices, allowing the model to capture spatial relationships and dependencies between different monitoring stations. A multi-channel

TCN is employed to extract temporal features, effectively capturing the temporal patterns and avoiding information leakage between time steps.

Attention mechanisms: The addition of attention mechanisms to various model architectures like CNNs, LSTMs, and even GANs has been a transformative development in DL. Attention allows models to focus on certain parts of the input when producing the output, making them more interpretable and capable of handling complex dependencies, including those found in spatiotemporal data. For example, Yu, Masrur, and Blaszcak-Boxe (2023) proposed SpatioTemporal (ST)-Transformer that utilizes a multi-head attention mechanism and a sparse attention mechanism to concentrate on the most useful contextual information across spatial, temporal, and variable-wise dimensions for improving spatiotemporal predictions of $PM_{2.5}$ concentrations in wildfire-prone areas. The integration of attention mechanisms into GNNs enhances the network's ability to focus on important nodes and edges when making predictions or representations. Focused on daily runoff modeling, Deng et al. (2023) integrated spatial graph-attention mechanisms with graph convolution layers. This allows the model to recognize both static and dynamic spatial patterns effectively. For wind power prediction at multiple locations and time steps, Wang and He (2022) employs graph attention networks (GATs) for capturing spatial dependencies between wind farms and uses a Transformer strategy for fusing multi-modal features, thus enabling the model to mine dependent information at various scales adaptively. Similarly, for wind power prediction, Chengqing et al. (2023) employed a deep graph attention reinforcement learning network that uses GAT for feature extraction and combines GRUs and TCNs for forecasting.

Others: Other novel modifications of existing model architectures have improved upon their ability to capture the dynamic spatial appearance of Earth system phenomena. Luo, Li, and Ye (2021) proposed a novel pseudo-flow spatiotemporal LSTM unit (PFST-LSTM), which includes a spatial memory cell and a position alignment module embedded in the structure of LSTM. The PFST-LSTM units are used in a new sequence-to-sequence architecture for precipitation nowcasting, effectively combining spatial appearances and motion information. Ravuri et al. (2021) introduced a deep generative model that can produce probabilistic predictions of precipitation from radar data, producing realistic and spatiotemporally consistent predictions over large regions and lead times up to 90 min ahead. The model utilizes a cGAN framework, incorporating spatial and temporal discriminators to ensure consistency in the prediction. The model introduces a regularization term to penalize deviations between real radar sequences and predictions, improving location accuracy. A fully convolutional latent module also allows predictions on larger spatial scales while maintaining spatiotemporal consistency.

5.4. Incorporating uncertainties

Reviewed models use various techniques to incorporate uncertainties and improve their prediction performance. Transfer learning, ensemble learning, and attention-based models are some of the techniques used by the models to address uncertainties. Hu et al. (2019) reduced dimensionality and estimated uncertainty in flood-induced conditions using proper orthogonal decomposition and prescriptive analytics with LSTM networks. Talib et al. (2021) compared two data-driven models (RF and LSTM) and a process-based land surface model for predicting daily evapotranspiration (ET) to comprehensively assess the presence of uncertainty in the model. The study emphasized the need for locally calibrated tools in areas where in situ ET, meteorological, and biophysical data are lacking, indicating uncertainty in the accuracy of the predictions and the need for further improvements in data availability and modeling techniques. Zhang et al. (2022) acknowledged the inherent unpredictability and complexity of wildfire events, as historical and existing spatiotemporal data can only provide limited insight into future fire occurrences. The authors evaluated the uncertainty of their DL model by comparing it to other approaches and

using evaluation metrics, such as the refined index of agreement, to assess the model’s performance. Tyson et al. (2023) incorporated uncertainties by addressing the high spatiotemporal variability of meteorological and hydrogeological processes in snow-dominated, karst watersheds. It used a physically based snow model to simulate snow processes and DL models to simulate streamflow based on the calculated snowmelt and potential evapotranspiration rates.

Additionally, several advanced statistical deep learning methods have been developed to better handle uncertainty in spatiotemporal forecasting. For example, Chengqing et al. (2023) used deep ensembles to capture model uncertainty, which involves training multiple instances of the same model with different initializations and combining their predictions. The variances among predictions of different models can provide a measure of uncertainty. Zammit-Mangion et al. (2022) employed approximate Bayesian methods, including variational inference, to make inferences with deep compositional spatial models, enhancing their ability to provide better predictions and uncertainty quantification. By optimizing a lower bound on the log-likelihood, variational inference provides a tractable way to estimate uncertainty in neural networks. Methods like quantile regression and probabilistic deep learning frameworks can predict a range of possible outcomes, providing a comprehensive measure of uncertainty (Nag, Sun, and Reich 2023; Ravuri et al. 2021). Ravuri et al. (2021) developed a deep generative model for probabilistic nowcasting of precipitation, which addresses challenges in capturing non-linear events and providing realistic spatiotemporal predictions. Nag, Sun, and Reich (2023) proposed a deep neural network-based model for spatiotemporal interpolation and forecasting, utilizing quantile-based loss functions to provide probabilistic forecasts. This method enhances the ability to account for non-Gaussianity and non-stationarity in the data, offering more comprehensive uncertainty quantification.

6. Advancements in spatiotemporal predictive applications

Table 4 summarizes the capabilities and advancements of DL applications in addressing key challenges in environmental and natural sciences – from climate modeling to environmental monitoring.

6.1. Climate modeling and weather prediction

Recent studies have demonstrated the efficacy of DL models in improving spatiotemporal prediction of various oceanic and atmospheric variables, including precipitation, sea ice concentration, sea surface height anomaly, and sea surface temperature. Table 5 synthesizes DL advancements specifically aimed at overcoming challenges in climate modeling and weather prediction.

Table 4. Capabilities of DL applications in addressing key challenges across domains.

Domain	Key challenges addressed	Prominent DL techniques	Real-world contributions
Climate modeling and weather prediction	Long-term forecasts, ensemble modeling	LSTMs, GANs, 1D-CNN	More accurate long-term forecasts, risk assessments
Disaster prediction and Management	Early warning, uncertainty modeling	CNNs, LSTMs, Autoencoders	Faster and more reliable early warning systems
Air quality prediction	Real-time monitoring, spatiotemporal variability	ConvLSTM, CNNs, Attention Mechanism	Real-time, fine-grained air quality forecasts
Hydrological modeling and prediction	Real-time flow, data scarcity	LSTM, GRU, XGBoost	Improved real-time flow prediction, flood risk assessments
Renewable energy forecasting	Wind behavior, temporal horizon	3D-CNN, Deep Convolutional Networks, DAGLN	Enhanced accuracy in wind and solar energy forecasts
Oceanography	Multidimensional data, eddy prediction	Attention Mechanism, ConvLSTM, GRU	Advanced ocean wave simulations, eddy trajectory prediction
Environmental monitoring	Water quality, continuous monitoring	ConvLSTM, Bidirectional LSTM, SHAP	Improved water quality parameters, continuous environmental monitoring

Table 5. Advancements in DL as applied to climate modeling and weather prediction.

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Data quality	CNNs; LSTM Attention Networks	Improved atmospheric phenomena identification; turbulence wavefront prediction	Quinting et al. (2022); Wu et al. (2022)
Non-linear behavior	LSTM Attention Networks; Autoencoders	Enhanced prediction of turbulence and climate indices like EQUINOO and ENSO	Wu et al. (2022); Saha and Nanjundiah (2020)
Spatiotemporal complexity	Sequence-to-Sequence Autoencoders; ConvLSTM	Advanced analysis of tropical cyclones; one-step-ahead land surface temperature prediction	Lian and Chen (2023); Kartal and Sekertekin (2022)
Computational demands	Encoder-Decoder Deep Residual Attention Network; Pseudo Flow Spatiotemporal LSTM	Improved computational efficiency in precipitation nowcasting	Che et al. (2022); Luo, Li, and Ye (2021)
Multi-source data integration	Multi-source Spatio-Temporal data fusion model (MUST)	Enhanced sea surface temperature prediction through data integration	Hou et al. (2022)
Operational utility	SAN-EFSModel; PrecipGAN	Real-world applications in tropical cyclone track prediction; high-resolution precipitation estimation	Dong et al. (2022); Wang, Tang, and Gentine (2021)
Increased spatiotemporal forecasting accuracy	Sequence-to-Sequence Autoencoders; CNNs	Improved accuracy in tropical cyclone and atmospheric phenomena prediction	Lian and Chen (2023); Quinting et al. (2022)
Model adaptability	Multi-source Spatio-Temporal data fusion model (MUST); LSTM Attention Networks	Integration of multiple data sources; adaptability to various environmental phenomena	Hou et al. (2022); Wu et al. (2022)
Real-time prediction	LSTM Attention Networks; Graph Convolutional Networks	Real-time adaptive optics correction; rare rainstorm prediction	Wu et al. (2022); Zhang et al. (2022)
Computational resource efficiency	Encoder-Decoder Deep Residual Attention Network (ED-DRAP)	Efficient resource usage in precipitation nowcasting	Che et al. (2022)

6.1.1. Advancements in predictive tasks

In the rapidly evolving field of climate modeling and weather prediction, recent studies have made significant strides in advancing our understanding and ability to predict various climate and atmospheric phenomena. CNNs have been effective in identifying specific atmospheric phenomena, such as warm conveyor belts, outperforming traditional trajectory-based methods (Quinting et al. 2022). An LSTM attention-based network, PredictionNet, demonstrated the ability to consider spatiotemporal coupling characteristics of turbulence wavefront (Wu et al. 2022). Sequence-to-sequence autoencoders have been employed to extract and analyze spatial appearance features and temporal dynamics of tropical cyclones from satellite images (Lian and Chen 2023). Autoencoders have also been effective in the prediction of critical climate indices like the EQUINOO and ENSO (Saha and Nanjundiah 2020). Hybrid models that combine ConvLSTMs have shown promising results in land surface temperature prediction (Kartal and Sekertekin 2022). GNN has tackled the challenge of predicting rare rainstorm events, addressing the issue of class imbalance (Zhang, Liu, and Wang 2022). Novel models such as the pseudo flow spatiotemporal LSTM unit (PFST-LSTM) and the encoder-decoder deep residual attention prediction network (ED-DRAP) have advanced precipitation nowcasting by effectively combining spatial and temporal features (Che et al. 2022; Luo, Li, and Ye 2021). GANs, like the PrecipGAN, have been useful for global satellite precipitation estimation at high spatiotemporal resolutions (Wang, Tang, and Gentine 2021). The development of SAN-EFSModel, an encoding-to-forecasting model based on ConvLSTM and spatial attention network, predicts tropical cyclone tracks by effectively extracting long-term spatial and temporal features (Dong et al. 2022).

6.1.2. Advancements in model's capability

The most significant improvement lies in the increased efficiency and accuracy of spatiotemporal forecasting (Lian and Chen 2023; Quinting et al. 2022; Wu et al. 2022). These advanced models, such as ConvLSTM, sequence-to-sequence autoencoders, and attention mechanisms, have also greatly enhanced the spatiotemporal understanding of environmental phenomena. The

autoencoder employed in predicting EQUINOO and ENSO indices (Saha and Nanjundiah 2020), as well as the ConvLSTM and MLP models for one-step-ahead land surface temperature prediction (Kartal and Sekertekin 2022), are instances of models that have effectively captured and learned from both spatial and temporal data. The adaptability of these models to different types of data and scales is another considerable advancement, as demonstrated across all the above-mentioned studies. The Multi-source Spatio-Temporal data fusion model (MUST) for predicting SST (Hou et al. 2022), in particular, showed its ability to integrate data from multiple sources with different spatial resolutions, enhancing the model's overall prediction capabilities and reliability. Recent studies have also improved the utility of DL models in real-time prediction and efficient resource usage, such as the real-time adaptive precipitation nowcasting (Che et al. 2022; Wu et al. 2022) and prediction of rare events like rare rainstorms (Zhang, Wang, and Liu 2022).

6.2. Disaster prediction and management

Recent studies improved disaster prediction and response, with enhanced accuracy in predicting earthquake magnitudes, road flooding status, lightning intensity, wildfire dangers, and burned areas, flood arrival times and peak discharges, and landslide displacements. Table 6 summarizes various challenges in disaster prediction and management and how advancements in DL have contributed to overcoming them.

6.2.1. Advancements in predictive tasks

The MyShake project has demonstrated a unique approach to building a smartphone seismic network for early earthquake warning through crowdsourcing (Kong, Martin-Short, and Allen 2020), introducing innovative simulation and clustering algorithms for global applicability. Another investigation examined spatial parameters to improve the prediction of earthquake magnitudes by introducing fault density in predicting high-magnitude earthquakes (Yousefzadeh, Hosseini, and Farnaghi 2021). A few studies have explored various models for predicting flood events. Fine-grained traffic data and spatiotemporal graph networks have been used for road segment flood status prediction (Yuan et al. 2022). An integrated LSTM and reduced order model framework has shown real-time predictive capability, potentially reducing casualties (Hu et al. 2019). Another study used ConvLSTM to significantly improve flood arrival times and peak discharge predictions (Chen et al. 2022). DL has been increasingly utilized to predict wildfire danger (Kondylatos et al. 2022) or burned areas of wildfires (Zhang, Liu, and Wang 2022). DL models based on graph convolution networks have shown high accuracy in predicting landslide displacement (Xi et al. 2023).

Table 6. Advancements in DL for disaster prediction and management.

Challenges & advancements	Deep learning techniques	Contribution to real-world applications	Reference studies
Earthquake prediction	Smartphone Seismic Network; SVM; DNN	Early earthquake warning; improved magnitude prediction based on new 'Fault Density' parameter	Kong, Martin-Short, and Allen (2020); Yousefzadeh, Hosseini, and Farnaghi (2021)
Flood prediction	STGCN; LSTM-ROM; ConvLSTM	Road network flood status; real-time flood prediction; improved arrival times and peak discharge prediction	Yuan et al. (2022); Hu et al. (2019); Chen et al. (2022)
Wildfire prediction	Temporal and Spatiotemporal DL models; CNN2D-LSTM	Improved wildfire danger prediction; advanced global burned area forecasting	Kondylatos et al. (2022); Zhang, Wang, and Liu (2022)
Landslide prediction	Attention-TGCN; Graph Convolution Networks	High spatiotemporal prediction accuracy for landslide displacement	Xi et al. (2023)
Spatiotemporal complexity	STGCN; Attention-TGCN	Enhanced spatial and temporal features in predicting flooding and landslides	Yuan et al. (2022); Xi et al. (2023)
Model capability and hybrid models	LSTM-ROM; CNN2D-LSTM	Real-time flood prediction; effective and improved wildfire forecasting	Hu et al. (2019); Zhang, Liu, and Wang (2022)

These demonstrate advancements in improving disaster predictions across earthquakes, floods, wildfires, and landslides, offering more reliable and real-time insights.

6.2.2. Advancements in model's capability

Similar advancements have been observed regarding DL model's capability, especially the improved ability to handle complex patterns and large datasets, as in the DNN model for earthquake magnitudes (Yousefzadeh, Hosseini, and Farnaghi 2021) and the LSTM and ROM integrated framework for real-time spatiotemporal flood predictions (Hu et al. 2019). Another advancement is the expanded inclusion of spatiotemporal features in DL models for more accurate and holistic predictions, such as including road networks in flood status prediction (Yuan et al. 2022) and the graph convolution networks to predict landslide displacement based on monitored time-series data (Xi et al. 2023). Another significant advancement lies in the combination of DL models to leverage the unique strengths of each. The CNN2D-LSTM model, for instance, combines the feature extraction capability of CNNs and the sequence learning ability of LSTMs, leading to effective and improved wildfire forecasting (Zhang, Wang, and Liu 2022). Another example can be seen in the proposed LSTM-ROM framework for spatiotemporal flood prediction, which combines the benefits of LSTM's temporal dependencies capturing ability and ROM's computational efficiency, making it effective for real-time applications (Hu et al. 2019). These hybrid models leverage the strengths of multiple algorithms, leading to improved speed and efficiency in predictive tasks.

6.3. Air quality prediction

The reviewed studies developed air quality prediction models that consider the need for accurate and representative data, as well as the need to account for the complexity of atmospheric processes and the interactions between different pollutants. Table 7 provides a quick overview of the DL techniques employed and their contributions to tackling various challenges in air quality prediction.

Table 7. Advancements in DL for air quality prediction.

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Predictive accuracy	Ensemble Algorithms (GBoost, CatBoost, XGBoost); Various DL Models	Improved accuracy in predicting benzene, PM2.5, and ground-level ozone concentrations	Hsu et al. (2022); Lyu et al. (2019); Wang et al. (2020)
Prediction granularity	Various DL Models	Hourly, daily, next 72 h, and continuous 24-hour air quality predictions	Faraji et al. (2022); Zhu et al., (2021); Xu et al. (2020); Mao et al. (2021)
Spatiotemporal analysis	Sequence-to-Sequence with Attention; ST-Transformer	Captured spatiotemporal correlations effectively; Enabled accurate predictions in different locations and timeframes	Jia, Cao, and Yang (2021); Wang et al. (2020); Yu et al. (2023)
Multi-pollutant Prediction	MMSL based on LSTM	Integrated different types of pollutant data; Captured complex interdependencies	Mao et al. (2021)
Data integration	Multiple ML algorithms; Satellite Data; CMAQ Simulations; LSTM	Combined different types of data for more comprehensive and accurate predictions	Hsu et al. (2022); He et al. (2022); Lyu et al. (2019); Li et al. (2017)
Model interpretability	Semi-SIDLM; ST-Transformer with Attention	Improved interpretability of air quality models; Enabled the analysis of influential factors during drastic changes	Zang et al. (2021); Yu et al. (2023)
Hybrid models	Res-GCN-BiLSTM; Sequence-to-Sequence with Attention; ResNet-ConvLSTM	Combined different types of neural networks; Enhanced feature extraction and prediction performance	Wu et al. (2023); Jia, Cao, and Yang (2021); Zhang et al. (2022)

6.3.1. Advancements in predictive tasks

Air quality prediction has seen several advancements in predictive accuracy, granularity, spatiotemporal analysis, and multi-pollutant predictions. Increased predictive accuracy has been achieved by implementing ensemble learning and DL models. For instance, ensemble algorithms (GBoost, CatBoost, and XGBoost) demonstrated improved predictive accuracy by capturing 92% of the variability in benzene concentrations (Hsu et al. 2022). Other studies exhibited superior performance in predicting $PM_{2.5}$ concentrations and ground-level ozone concentrations through DL models (Lyu et al. 2019; Wang et al. 2020). Advancements have also been made in predicting air quality at finer granularities, such as hourly and daily predictions (Faraji et al. 2022), next 72 h predictions (Zhu et al. 2021), continuous 24-hour predictions at specific stations and regional levels (Xu et al. 2020), and predicting the next 24 h average $PM_{2.5}$ concentration (Mao et al. 2021). Various studies have effectively captured spatiotemporal correlations in air quality, meteorological, and emission data. Sequence-to-sequence models with attention mechanisms (Jia, Cao, and Yang 2021; Wang et al. 2020) and the SpatioTemporal (ST)-Transformer (Yu et al. 2023) have shown particular promise in spatiotemporal predictions. Multi-pollutant prediction has been proposed, focusing on capturing the interactions and correlations between different pollutants. Mao et al. (2021) introduced the Multi-Output and Multi-Index Supervised Learning (MMSL) model, based on LSTM, which integrates air pollution, meteorological, and gaseous pollutant data, enabling the capture of intricate interdependencies among different pollutants.

6.3.2. Advancements in model's capability

Several models have shown the ability to integrate different types of data for air quality predictions, such as assessing discrepancies between satellite observations and bottom-up inventories of nitrogen oxide (NO_x) emissions (He et al. 2022), and fusing CMAQ simulations with ground-level observations (Lyu et al. 2019). Some models have advanced in model interpretability, such as investigating ozone spatiotemporal characteristics using a tree-based ensemble model (Zang et al. 2021), and leveraging attention mechanisms to distinguish wildfire impacts on air quality (Yu, Masrur, and Blaszcak-Boxe 2023). Several studies have proposed hybrid models that combine different types of neural networks. For example, the Res-GCN-BiLSTM model combined a residual neural network, a graph convolutional network, and a bidirectional long short-term memory for short-term regional NO₂ and O₃ concentration prediction (Wu et al. 2023). Jia, Cao, and Yang (2021) proposed another hybrid model that combined a sequence-to-sequence model with attention mechanisms for predicting ground-level ozone concentrations. The sequence-to-sequence model was effective in capturing the temporal dependencies in the data, while the attention mechanism helped the model focus on important time steps. In Zhang et al. (2022), a hybrid model combining a ResNet and a ConvLSTM network was developed for $PM_{2.5}$ prediction. ResNet was used to extract spatial distribution features, while ConvLSTM was used to extract spatiotemporal features. These studies highlighted the utility of hybrid models in effectively extracting different types of features from the data to enhance prediction performance.

6.4. Hydrological modeling and prediction

The articles discussed employ DL models to tackle key challenges in hydrology such as high spatiotemporal variability, nonlinearity, accurate groundwater level prediction in high-demand areas, uneven groundwater distribution in arid regions, and accurate streamflow simulation in diverse basins. Table 8 offers a quick overview of the state-of-the-art DL techniques in hydrological modeling and prediction, detailing their applications and contributions to addressing various challenges.

6.4.1. Advancements in predictive tasks

Multifactorial predictions in hydrology such as predicting groundwater levels considering spatiotemporal and multidepth factors, have been effectively addressed using SVM and FFNN models

Table 8. Advancements in DL for hydrology modeling and prediction.

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Multifactorial predictions	SVM and FFNN models	Predicting groundwater levels considering spatiotemporal and multidepth factors; Handling complex interactions and non-linear relationships	Malakar et al. (2023)
Spatial predictions	Spatiotemporal graph convolution	Addressed non-Euclidean topological river networks; Improved spatial accuracy in hydrological predictions	Deng et al. (2023)
Complex hydrological processes	Physically based snow model combined with ConvLSTM	Simulated interactions between snowmelt, rainfall, and streamflow in challenging terrains like snow-dominated, karst basins	Tyson et al. (2023)
Varied spatiotemporal scales	Various DL Models (LSTM-CNN hybrids, RF, DNN, CNN)	Improved predictions in large transboundary river basins and arid regions; Daily runoff predictions across non-Euclidean topological river networks	Malakar et al. (2023); Wang et al. (2022); Li et al. (2022)
Long-term Predictions	Deep CNN combined with a Transformer	Enhanced 12-month ahead streamflow forecasts; Mitigated limitations of traditional models	Ghobadi and Kang (2022)
Increased predictive accuracy	CNN, LSTM-CNN hybrids, 3D-CNN combined with Transformer	Outperformed traditional models like RF and SWAT; High accuracy in long-term streamflow prediction	Wang et al. (2022); Li et al. (2022); Ghobadi and Kang (2022)

(Malakar et al. 2023). These models can handle the complex interactions and non-linear relationships among various hydrological variables. Spatial predictions involving non-Euclidean topological river networks, which pose challenges for traditional hydrological models, have seen promising advancements with the use of a spatiotemporal graph convolution model (Deng et al. 2023). This novel model is adept at handling the complexities of non-Euclidean spatial structures, contributing to more accurate and efficient hydrological predictions. Complex hydrological processes, like the interactions between snowmelt, rainfall, and streamflow in snow-dominated, karst basins, have been effectively simulated by combining a physically based snow model with a DL model (Tyson et al. 2023). This innovative approach has overcome some of the limitations of traditional models and has shown potential for better streamflow simulations in such challenging terrains.

6.4.2. Advancements in model’s capability

These advanced techniques have enabled prediction at varied spatial and temporal scales. For instance, the prediction of groundwater levels in the large transboundary Indus-Ganges-Brahmaputra-Meghna river basin (Malakar et al. 2023) and the prediction of groundwater potential in the vast Qaidam Basin (Wang et al. 2022) illustrate this capability. The spatiotemporal graph convolution-based model has even demonstrated its ability to accurately predict daily runoff across non-Euclidean topological river networks (Deng et al. 2023). For long-term predictions like 12-month ahead streamflow forecasts, traditional models often fall short due to the dynamic and non-linear nature of hydrological processes. This has been mitigated by employing an integrated model that combines a deep CNN with a Transformer (Ghobadi and Kang 2022). The reviewed models have also resulted in increased predictive accuracy. The LSTM-CNN hybrid model for river flow prediction in large river basins surpassed the predictive performance of the traditional Soil and Water Assessment Tool model, particularly in wet seasons (Li et al. 2022).

6.5. Renewable energy forecasting

The challenges of wind power forecasting, such as the complexity of forecasting, temporal horizon, and physical consistency, have been significantly addressed through recent advancements. [Table 9](#)

Table 9. Advancements in DL for renewable energy forecasting (wind power).

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Wind behavior forecasting complexity	3D-CNN, Deep Convolutional Networks, DAGLN	Increased multi-step prediction accuracy; Handling spatiotemporal behavior; Outperforming state-of-the-art models in offshore wind power predictions	Zhu et al. (2021); Yu et al. (2019); Ren et al. (2023)
Temporal horizon challenge	M2STAN (Multi-Modal Multi-Task Spatiotemporal Attention Network), SPLNet (Time-variant structure)	Ultra-short-term predictions; Improved modeling of wind speed's volatile nature across multiple time steps	Wang and He (2022); Ye et al. (2022)
Hybrid model capability	Ensemble of GAT, GRU, TCN; GCN combined with LSTM	Improved forecasting performance by integrating multiple models to handle complex spatiotemporal data	Chengqing et al. (2023); Liao et al. (2022)
Physical consistency	Physics-informed DL techniques, Navier-Stokes equations	Improved model reliability and interpretability by ensuring physical consistency in spatiotemporal wind fields	Zhang and Zhao (2021)

summarizes advancements in DL for renewable energy forecasting, specifically focusing on wind power forecasting challenges and the techniques applied to address them.

6.5.1. Advancements in predictive tasks

The forecasting complexity of wind behavior relates to intricate, interdependent factors affecting wind patterns and the non-linear, non-stationary nature of wind speed data. Several studies have proposed models to handle these complexities, including a deep convolutional network approach to focus on spatiotemporal dynamics of wind behavior (Yu et al. 2019), a 3D-CNN-based model that uses historical spatiotemporal behavior features to increase the accuracy of multi-step wind speed predictions (Zhu et al. 2021), and incorporating the spatiotemporal correlations of multi-node wind speed data through data-driven graph construction and graph learning for offshore wind power prediction (Ren et al. 2023). The temporal horizon challenge relates to the difficulty in making accurate predictions over various time scales. The multi-location ultra-short-term wind power predictions have been achieved using a graph attention network and bidirectional gated recurrent unit to model spatial and temporal dependence (Wang and He 2022). Ye et al. (2022) incorporated a time-variant structure in their model to improve the modeling of wind speed's volatile nature across multiple time steps.

6.5.2. Advancements in model's capability

Hybrid architectures have been developed to leverage the strengths of different model types and integrate them to handle complex wind power data, such as the ensemble of GAT, GRU, and TCN for wind power forecasting (Chengqing et al. 2023) and the combination of GCN and LSTM to capture spatiotemporal features simultaneously (Liao et al. 2022). Maintaining physical consistency has been enhanced within predictive models. Zhang and Zhao (2021) incorporated the Navier-Stokes equations into a deep neural network, combining LIDAR wind speed measurements at sparse locations and flow physics to predict spatiotemporal wind fields.

6.6. Oceanography

The reviewed studies focus on applying DL techniques to various aspects of oceanographic research, including eddy trajectory prediction, 3D ocean temperature prediction, ocean wave simulation, and sea clutter parameter inversion. Table 10 summarizes advancements in DL for oceanographic research, focusing on the challenges and the techniques applied to address them.

Table 10. Advancements in DL for oceanographic research.

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Mesoscale eddy trajectory prediction	Attention-based DL, CSA-encoder, TA-GRU	Enhanced accuracy in eddy trajectory predictions; Superior feature extraction from multidimensional eddy data	Wang, Wang, and Han (2022)
3d ocean temperature prediction	Conv1D, LSTM	Improved modeling of temporal dependencies and dynamic patterns in 3D ocean temperatures	Xiao et al. (2022)
Ocean wave simulation	ConvLSTM	Real-time, authentic ocean wave simulation; outperforms traditional FFT methods	Li et al. (2022)
Ocean current prediction	Spatiotemporal Coupled Attention Networks, STCANet	Superior extraction of temporal and spatial coupling information, outperforming traditional methods like ARIMA, LSTM, GRU, and CNN_GRU	Xie et al. (2023)
Sea clutter parameter inversion	LSTM, DNN	Improved radar detection and remote sensing through enhanced sea clutter prediction and parameter inversion	Linghu et al. (2023)
Enhanced model capability	Attention Mechanisms, Advanced Architectures like Conv1D, ConvLSTM, STCANet	More targeted and precise predictions due to attention mechanisms; capturing complex spatiotemporal dependencies	Wang et al. (2022); Xiao et al. (2022); Li et al. (2022); Xie et al. (2023)

6.6.1. Advancements in predictive tasks

The prediction of mesoscale eddy trajectories has been enhanced using an attention-based model by effectively extracting aggregated features from multidimensional eddy data (Wang et al. 2022), where traditional models have struggled with efficiently managing large-sized data instances and pinpointing critical eddy characteristics. Focused on predicting the long lead dynamics of the three-dimensional ocean temperature, Xiao et al. (2022) incorporated Conv1D and LSTM layers to model the temporal dependencies and dynamic patterns at different spatial locations and depths. ConvLSTM has been employed to accelerate the wave simulation process, improving the realism of the simulations, enabling the prediction of future wave height fields, and outperforming traditional methods such as the fast Fourier transform (FFT) algorithm in real-time performance (Li et al. 2022). The emergence of spatiotemporal coupled attention networks has enabled the traditionally challenging extraction of temporal and spatial coupling information about the behavior of ocean currents (Xie et al. 2023). The prediction of sea clutter characteristics and parameter inversion has been advanced using a combined LSTM and DNN model that established correlations between marine environmental elements and sea clutter characteristics (Linghu et al. 2023), demonstrating a significant improvement in sea surface remote sensing and sea clutter suppression.

6.6.2. Advancements in model's capability

Advanced architecture and attention mechanisms have been employed to enhance model capabilities in various aspects. In Wang et al. (2022), the capability of their spatiotemporal attention-based architecture for mesoscale eddy trajectory predictions was achieved by combining a novel autoencoder, equipped with channel and spatial attention mechanisms and a GRU network supplemented with a temporal attention layer. In 3d ocean temperature prediction, latent features and long-term temporal dependencies were extracted by incorporating a Conv1D and two LSTM layers. By substituting the traditional FFT method with a ConvLSTM-based model, Li et al. (2022) accelerated the wave simulation process and enhanced the prediction of future wave height fields. Spatiotemporal coupling characteristics of ocean currents have been extracted using a spatiotemporal coupled attention deep network model that combined a spatial module and a spatial channel attention module (Xie et al. 2023).

Table 11. Advancements in DL for environmental monitoring.

Challenges & advancements	DL techniques	Contribution to real-world applications	Reference studies
Harmful algal blooms prediction	ConvLSTM, Bidirectional LSTM	Enhanced capability in forecasting harmful algal blooms (HABs), providing valuable insights into their dynamics	Tang et al. (2022); Ni et al. (2022)
NDVI from satellite data	Deep Stacking Network (Deep-STEP)	Spatiotemporal prediction of NDVI from satellite remote sensing data	Das and Ghosh (2016)
Continuous monitoring & prediction	DNN	Continuous monitoring crucial for managing and preventing environmental degradation	Ramaraj and Sivakumar (2023)
Enhanced model capability	Multisource Hybrid Datasets, Attention Mechanism	Improved prediction accuracy and understanding of complex systems by integrating multi-source data and applying attention mechanisms	Tang et al. (2022); Ni et al. (2022); Ramaraj and Sivakumar (2023)
Model interpretability	SHAP, Attention Mechanism	Enhanced interpretability allowing for a better understanding of significant variables affecting water quality	Zheng et al. (2023); Ni et al. (2022)

6.7. Environmental monitoring

The reviewed studies focus on using DL methods to predict or estimate water quality parameters, harmful algal blooms, and cyanobacterial blooms in lakes. Table 11 offers an overview of the latest advancements in DL techniques used for environmental monitoring.

6.7.1. Advancements in predictive tasks

The incorporation of models capable of handling spatiotemporal predictions and predicting complex phenomena has been a significant improvement. The ConvLSTM employed in Tang et al. (2022) and the bidirectional LSTM network in Ni et al. (2022) have demonstrated strong capabilities in predicting temporal and spatial variations, providing valuable insights into the dynamics of phenomena such as harmful algal blooms and chlorophyll-a concentrations. Similarly, the Deep-STEP presented by Das and Ghosh (2016), which employs a deep stacking network for spatiotemporal prediction of NDVI from satellite remote sensing data, provides valuable insights into spatial features over time. Moreover, DL has empowered us to integrate multi-source data for improved predictions. Tang et al. (2022), for instance, fused remote sensing data with physics-based modeling, resulting in a comprehensive understanding of the system. This hybrid approach not only improves prediction accuracy but also addresses the limitations of individual datasets. The capability for continuous monitoring and prediction, as demonstrated in Ramaraj and Sivakumar (2023), is crucial in managing and preventing environmental degradation.

6.7.2. Advancements in model's capability

The reviewed models have demonstrated high accuracy in predicting water quality parameters, with promising performance for short – and mid-term forecasting. The use of multisource hybrid datasets that fuse remote sensing with physics-based modeling has improved the performance of the models significantly. Interpretable DL methodologies, such as the SHAP method in Zheng et al. (2023), have enhanced our ability to identify significant variables and understand their contributions to predictions, providing an opportunity to deepen our grasp of the factors and processes driving the changes in water quality. In addition, Ni et al. (2022) utilized an attention mechanism to calculate the weights for the characteristic factors affecting chlorophyll-a concentration, offering insights into their relative importance and interaction dynamics. While DL does not directly provide causal relationships or a deep understanding of physical principles, when used alongside traditional physics-based modeling, as illustrated in Tang et al. (2022), it can enhance the simulation

speed and prediction performance. This synergy between traditional and DL methodologies could be a promising avenue for improving our understanding of complex physical systems.

7. Gaps and future research

7.1. Interpretability and explainability

The reviewed studies have achieved notable progress in model interpretability, leveraging techniques such as attention mechanisms, graph networks, and SHAP to elucidate the contributions of specific inputs and features to the model's predictions. However, they also acknowledge the inherent challenges due to the complexity of the models and the high-dimensional, temporal nature of the data, and highlight the need for more comprehensive interpretation techniques and improved communication of their implications.

7.1.1. Complex Neural Networks, Interpretability Tools, and Model Complexity

- Model complexity:** Complex neural networks, especially those that combine separate spatial and temporal modules, are non-linear and high-dimensional, making them hard to interpret – their intricate architecture with thousands to millions of parameters combined with layers of non-linear transformations. The connections between the input and output are hidden, making the prediction process challenging to understand and interpret. Several tools and techniques, including Layer-wise Relevance Propagation, Local Interpretable Model-agnostic Explanations (LIME), Integrated Gradients, and SHAP, have all been developed to provide insights into these complex models. LIME, for instance, can provide localized interpretability by generating simpler, more intuitive models around a prediction, offering a clearer understanding of the model's decisions. Similarly, libraries like Captum offer various attribution methods highlighting the importance of input features and their contribution to the outputs. Furthermore, SHAP unifies several existing methods to assign a cohesive measure of feature importance for individual predictions. SHAP values denote not only the importance of features but also the magnitude and direction of each feature's effect. While these tools significantly enhance the interpretability of complex models, they each provide model interpretation to a certain degree. They might elucidate the impact of individual features but may not comprehensively represent complex feature interactions. In addition, these tools are not explicitly designed to handle high-dimensional, temporal data, which is the typical case for spatiotemporal forecasting. **Future research** can focus on developing advanced interpretability tools designed specifically for handling complex, high-dimensional data in spatiotemporal forecasting, therefore enhancing the comprehensibility of complex feature interactions and model behaviors.
- Trade-off between Accuracy and Interpretability:** Challenge remains to the trade-off between predictive accuracy and model interpretability. Highly complex models, such as DL models, can typically capture intricate patterns in data and yield superior predictions. However, their complex architectures and huge parameter space become significantly challenging to interpretability. On the other hand, simpler models, such as the decision trees used by Zang et al. (2021), are highly interpretable due to their straightforward decision-making process, but may not capture complex patterns in the data as effectively as their complex counterparts. **Future research** should develop new frameworks and methodologies to balance accuracy and interpretability, making complex models more understandable. Also, there is a need for more investigation into techniques that ensure simpler models, like decision trees, capture complex patterns more effectively.

7.1.2. Understanding Decision-Making Process:

- **Explainability vs. Interpretability:** Techniques like attention mechanisms offer explainability by displaying which features a model considers important, but they do not necessarily reveal a comprehensive understanding of the decision-making process. There remains a gap between explainability (what the model pays attention to) and interpretability (why and how the model makes specific decisions). *Future research* needs to focus on developing new techniques to not only highlight the importance of certain features, but also explain how and why these features affect the model's decision-making process.
- **Prediction Errors, Biases, and Model Reliability:** Prediction errors and biases can significantly impact the interpretability of models. When a model's predictions are consistently biased or erroneous, it can lead to uncertainty about its outputs (Lyu et al. 2019). This uncertainty is not merely a technical problem but also influences interpretability because it can raise questions about the model's reliability. If a model's predictions are unreliable, interpreting the model's decision-making process becomes challenging because the credibility of its decisions is in doubt. This means that an integral part of improving model interpretability is not just understanding the model's decision-making process, but also improving its accuracy and reducing its biases. *Future research* should enhance the reliability of these models, subsequently leading to improved model interpretability, as predictions would be more credible.
- **Understanding Predictor Fields:** Understanding the impact and interactions of various input features or predictor fields is crucial for model interpretability. Quinting et al. (2022) emphasized the need to examine predictor fields that might be causing a deterioration in a CNN model's reliability. This problem is compounded when dealing with high-dimensional data, where there can be a large number of predictor fields with complex interdependencies. *Future research* should focus on developing techniques to better understand the impact and interaction of predictor fields, especially in models dealing with high-dimensional data, for a more profound understanding of the decision-making process in complex models.

7.1.3. Lack of Explicit Process Representation and Context-Specific Interpretations:

- **Lack of Explicit Process Representation:** DL models in ESS often have to deal with complex real-world processes, but these processes might not be explicitly represented within the models. Talib et al. (2021) pointed out the absence of root zone water dynamics in DL models, making models less interpretable and more challenging to explain. Without explicitly modeling known physical or biological processes, it becomes difficult to understand how the model approximates these processes or how the changes in these processes over time might impact the model's output. *Future research* should aim to develop DL models that can more explicitly incorporate known physical or biological processes, enhancing their interpretability. Researchers can explore ways to integrate domain knowledge into DL models to create hybrid models that blend data-driven learning with known process dynamics.
- **Context-Specific Interpretations:** Model interpretability is often context-dependent, meaning that models might be easily interpreted within a specific domain or context they are trained for but become opaque when applied elsewhere. This notion is echoed by Talib et al. (2021), who discussed the issue of local calibration. The complexity increases when there is a need to generalize the model across different regions, time periods, and situations (Dun, Yang, and Lei 2022; Zang et al. 2021; Zhu et al. 2021). Models that perform optimally within one context may not yield similar results in another, leading to difficulties in interpreting and explaining their behavior across varying contexts. *Future research* should develop techniques to improve model generalizability while maintaining transparency and investigate how interpretability changes across different contexts and how to minimize such effects.

7.2. Integration with physical models

Explicit Integration: Explicit integration – incorporating physical principles or equations directly into DL models – is largely absent. Only one of the reviewed studies incorporates the Navier-Stokes equations (Zhang and Zhao 2021), a set of physical equations describing fluid dynamics based on principles of momentum conservation, into a DL model to provide insights into wind speed forecasting. The absence of explicit integration with established physical models of ESS is a substantial gap. Challenges remain in the complexity of the physical equations, the difficulty in translating these equations into a form that can be incorporated into DL models, or the lack of methods for effectively combining these two fundamentally different modeling approaches. **Future research** needs to invest in developing robust methodologies to incorporate physical equations into DL models.

Correcting Model Misspecification in Physics-Informed Neural Networks (PINNs): Despite the effectiveness of PINNs for discovering governing equations, the physical models encoded in PINNs may be misspecified in complex systems as some of the physical processes may not be fully understood, leading to poor accuracy in PINN predictions. Zou, Meng, and Karniadakis (2024) presented a general approach to correct the misspecified physical models in PINNs for discovering governing equations, given some sparse and/or noisy data. By encoding the assumed physical models in PINNs and employing additional deep neural networks (DNNs) to model the discrepancy between the imperfect models and the observational data, the proposed method reduces computational errors caused by model misspecification. Furthermore, Bayesian PINNs (B-PINNs) and ensemble PINNs can provide reasonable uncertainty bounds in the discovered physical models, enhancing the reliability of predictions. This approach is particularly beneficial in complex systems where the physical processes are not exactly known. **Future research** should explore this line of methodology further to improve the integration of physical models into DL frameworks, ensuring more accurate and reliable predictions.

Compatibility and Harmonious Integration: Ensuring that DL models are compatible with physical models in terms of data inputs, outputs, and general behavior is a significant challenge and research gap. Li et al. (2021) highlighted the need to test the transfer learning method with other datasets and integrate it with land surface models for soil moisture prediction. Wang et al. (2020) emphasized the need to integrate DL models with carbon capture and storage technologies, and the integration process must consider the inherent characteristics and limitations of both the DL and physical models to prevent the drawbacks of one from negatively impacting the performance of the other. Similarly, in Lyu et al. (2019), the fusion product derived from Chemical Transport Model (CTM) simulations and ground observations via a DL approach provides full coverage without sampling biases, but it suffers from modeling biases due to the limitations of the CTM simulations and the imperfections in the modeling framework itself. **Future research** can tackle the challenge to ensure that DL models are compatible with physical models in terms of data inputs, outputs, and general behavior, and that the limitations of one do not negatively impact the other.

Modeling of Complex Physical Phenomena: Applying DL techniques to understand and forecast complex physical phenomena is challenging. Saha and Nanjundiah (2020) highlight the application of LSTM networks to study and predict climate indices based on past time-series data. Quinting et al. (2022) focus on further investigation of biases in WCBs occurrence frequency over the North Atlantic using operational ensemble forecasts. Such complex physical phenomena often have nonlinear and multifactorial influences, making their prediction and understanding challenging. Physical models provide a deterministic representation of these phenomena based on known physical laws, which can help capture their essence. However, these models may not adequately

account for stochasticity or unexpected variations in these phenomena. On the other hand, DL models can capture these nonlinear patterns and variations from the data but may not incorporate the physical laws governing these phenomena. Therefore, effectively integrating these two types of models requires a deep understanding of the physical phenomena, an adequate representation of the physical laws in the DL model, and the capability to handle the stochastic nature of these phenomena. Overcoming this challenge can lead to more accurate and robust forecasts, contributing to a deeper understanding of complex environmental systems. **Future research** can focus on how to develop hybrid models that effectively combine the strengths of DL and physical models for studying complex physical phenomena.

Causal Understanding: While DL can provide insights and correlations, it does not directly provide causal relationships or explain the physical principles underlying the ESS system. Moreover, a deep understanding of causality can also aid in model generalization, leading to accurate predictions even in unseen or changing scenarios where the observed patterns differ from the training data. **Future research** should explore ways to fill in the gap in the current capabilities of DL models when it comes to causal understanding. Bridging this gap would improve the interpretability of these models and potentially enhance their robustness and predictive accuracy, especially in the context of physical systems where causal relationships are important.

7.3. Generalizability and scalability

The gap in generalizability and scalability lies in the need for more comprehensive studies that validate the effectiveness of DL models, especially those using transfer learning methods, across different datasets, contexts, and scales.

Challenges in Generalizability and Scalability: The reviewed studies highlight the difficulty in generalizing DL models to different contexts (Hsu et al. 2022; Li et al. 2021; Talib et al. 2021; Wang, Tang, and Gentine 2021; Zang et al. 2021; Zhu et al. 2021). When applying the model to other geographical locations, extrapolating to unobserved conditions or extreme events, or scaling to different regions and time periods, the models often struggle to maintain their accuracy. **Future research** should aim to develop models or adapt existing strategies, such as domain adaptation, data augmentation, multi-modal learning, and few-shot learning, with broader spatial and temporal applicability while maintaining consistent performance levels.

Limited Validation of Transfer Learning: The research done by Li et al. (2021) and Lu et al. (2022) validates the effectiveness of transfer learning in two specific applications: soil moisture prediction and lightning intensity estimation respectively. In both cases, models pretrained on large datasets (ERA5-land and large image datasets respectively) were fine-tuned on specific task-related datasets (SMAP dataset and lightning intensity data), yielding promising results. However, these individual studies do not necessarily guarantee a broad applicability of transfer learning in ESS. In addition, the reviewed papers have yet to explore the potential benefits and limitations of self-supervised learning methods, which could offer alternative ways to leverage large, unlabeled datasets for model training. **Future research** is needed to test the efficacy of transfer learning and self-supervised learning across a broader range of datasets, tasks, and application areas within ESS. It also involves identifying the specific conditions or factors that affect the success of transfer learning and developing strategies to manage these factors.

Absence of Comprehensive Comparative Studies: Various methodologies and approaches, including self-supervised learning, zero-shot learning, meta-learning, and domain adaptation, exist to augment model generalizability. However, there is a lack of comprehensive studies comparing different transfer learning strategies in the domain of ESS. Such comparative studies

are essential to understanding the relative merits and demerits of different strategies and to provide guidelines for selecting appropriate strategies for specific applications. Correspondingly, *future research* efforts could pay attention to exploring and documenting diverse strategies for adapting models to new datasets, regions, tasks, and domains while optimizing their performance.

7.4. Uncertainty quantification

Lack of Methods for Handling Prediction Errors: As Lyu et al. (2019) mentioned, even though analyses at the regional level can mitigate uncertainties, there still exist uncertainties arising from prediction errors, as small errors can propagate and significantly affect the final prediction. Current approaches, such as cross-validation, bootstrapping, or ensemble methods, are not always sufficient or appropriate to quantify and manage these errors in DL models. Developing and incorporating robust methods for quantifying prediction errors will provide accurate measures of error and aid in understanding the sources and types of these errors. Techniques such as Monte Carlo Dropout (Gal and Ghahramani 2016), deep ensembles (Lakshminarayanan, Pritzel, and Blundell 2017), and uncertainty-aware transfer learning (Fischer, Orescanin, and Eckstrand 2024) offer promising avenues for improving error quantification. *Future research* could leverage these methods to handle the complexities and unique challenges of ESS data, such as spatial and temporal autocorrelation, non-linearity, and heterogeneity.

Complex Natural Processes Representation: Complex natural processes have intricated, often non-linear, relationships that are influenced by various environmental variables. A failure to adequately capture these complexities can lead to inaccurate model predictions, creating uncertainty in model outcomes. Talib et al. (2021) noted that models lack representation of root zone water dynamics, leading to uncertainty in model outcomes. *Future research* should strive to bridge this gap in accurately incorporating and representing intricate natural processes within models, such as using physics-informed neural networks (Kashinath et al. 2021) that integrate physical laws into deep learning models to enhance their ability to capture complex environmental dynamics.

Integration of Advanced Techniques for Uncertainty Quantification: The integration of advanced techniques for uncertainty quantification in ESS models, like using Bayesian Neural Networks (BNNs), remains a significant challenge (He et al. 2022). BNNs provide a probabilistic framework, producing a prediction and a measure of uncertainty related to that prediction. However, incorporating BNNs into models, particularly in combination with existing architectures like the U-net model, might not be able to benefit from the strengths of both models – the U-net model's capability in handling spatial data and BNN's capability in uncertainty quantification. In addition to BNNs, *future research* endeavors could also explore and integrate additional uncertainty quantification techniques, such as variational inference (Fischer, Orescanin, and Eckstrand 2024), deep Gaussian processes (Damianou and Lawrence 2013), and spatio-temporal deep Kriging (Nag, Sun, and Reich 2023), making them more reliable for ESS applications. For example, Psaros et al. (2023) propose a comprehensive framework for integrating uncertainty quantification into PINNs, neural operators, and stochastic partial differential equations solvers, which includes posterior inference, prior learning, data noise modeling, and post-training calibration. This framework could be instrumental in addressing the diverse problems in scientific machine learning. Additionally, Meng et al. (2022) developed a new Bayesian framework based on deep neural networks to extrapolate in space-time using historical data and quantify uncertainties arising from both noisy and gappy data. This approach employs physics-informed Generative Adversarial Networks (PI-GAN) to learn functional priors and utilizes Hamiltonian Monte Carlo (HMC) for posterior estimation, effectively integrating data from multiple fields and providing accurate predictions as well as uncertainty quantification.

Handling Missing Data: Faraji et al. (2022) highlighted the challenge of managing missing data, particularly in time series, which presents a significant gap in ESS DL models. Missing data disrupts the sequence and impacts long-term prediction accuracy, increasing uncertainty. Techniques such as data imputation using GANs (Yoon, Jordon, and Schaar 2018) or integrating temporal convolutional networks (TCNs) (Yu, Yin, and Zhu 2018) can provide effective solutions for handling missing data. **Future research** should develop more robust strategies to handle missing data effectively, estimate missing values based on existing data trends, and incorporate the induced uncertainty into final predictions.

Prediction of Extreme Values: ESS often involves challenging predicting extreme events or conditions (Jia, Cao, and Yang 2021; Mao et al. 2021). The primary difficulty stems from the fact that these extreme events are rare and thus represent a small portion of the data available for training. This imbalance in the training data can bias the model towards more common, less extreme events and conditions, resulting in poor performance when predicting these crucial extremes. Moreover, the uncertainty associated with the predictions is high as extreme events often lie outside the range of conditions seen in the training data, adding a layer of complexity to the problem. Approaches such as synthetic minority over-sampling (SMOTE) (Chawla et al. 2002) combined with uncertainty quantification techniques like Bayesian deep learning (Wilson and Izmailov 2022) can enhance the ability to predict and quantify uncertainties in extreme events. Therefore, **future research** should incorporate more advanced techniques to handle sample imbalance and effectively quantify and incorporate the inherent uncertainty when predicting these extreme events.

8. Conclusion

This review serves as a comprehensive demonstration of the advancing development of DL models in ESS across various application fields, from climate modeling to disaster management. The review highlights general innovations in the novel development of DL models and advancements in application capabilities, especially how these works address the challenges specific to ESS, such as spatio-temporal dependence, scale dependencies, and the urgent need for model generalizability. Looking ahead, three primary directions for future research have emerged. First, the development of more interpretable hybrid models promises a synergistic fusion between deep learning and traditional physical models, potentially yielding more accurate and understandable forecasting approaches. Second, the concept of model generalizability will gain increasing importance, with techniques like domain adaptation and transfer learning offering promising pathways to make models more versatile and applicable across varying geographic settings. Finally, as ESS phenomena are inherently uncertain, methods for robust uncertainty quantification and handling missing data will be critical in making these predictive models not just accurate, but also reliable and accountable.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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